DRAFT

STAMPEDE/BLUEWAY

SPI Interface Specification

The STAMPEDE/BLUEWAY BFIC has a digital SPI interface and dedicated control pins for configuring, controlling, and monitoring the chip. The document will describe the digital interface and provide a guide for programming the chip.

SPI Overview

The SPI interface is a serial programming interface for the STAMPEDE/BLUEWAY beamforming ICs. It supports reading and writing configuration, status, and control registers through a 1-bit serial data interface.

Features:

- Read or Write of 16-bit registers with a single 32-bit transaction.
- Continuous sequential reading or writing of registers through payload extension.
- Continuous sequential writing of beam index pointers through payload extension.
- 100 MHz serial clock.
- 8-bit CHIP ID to allow for individual addressing of up to 255 devices on a single data line.
- Auto-enumeration or pin programming of CHIP ID
- Broadcast addressing to write all devices simultaneously.
- Indirect addressing of registers through an address pointer.
- Auto-increment of address pointer for simplified accessing of sequential registers.
- 1024 Beam table sets.
- 2 beams per channel
- Asynchronous beam table index update.
- Enable state machine to manage LNA power and switch state.
- ADC capture control for sampling power detector and temp sensor
- ADC clock control – programmable divider and enable for ADC clock.
- Efuse based calibration of bias currents, temp sensors, power detectors, and RTPS.
- 32-bit unique ID
- Efuse programming and readback interface
- Output buffers for Daisy chain connection of devices

STAMPEDE/BLUEWAY Theory of Operation

The STAMPEDE/BLUEWAY device is a beam forming IC with 8 channels of transmit or receive processing grouped into 2 I/O ports as show in the figure below. It supports two independent beamformer phase and gain control for each channel. It includes a common mode gain adjustment for each I/O port. Transmit power and temperature telemetry can be read though the SPI control interface. Embedded efuse can be programmed so that the device will initialize calibration and bias after reset.

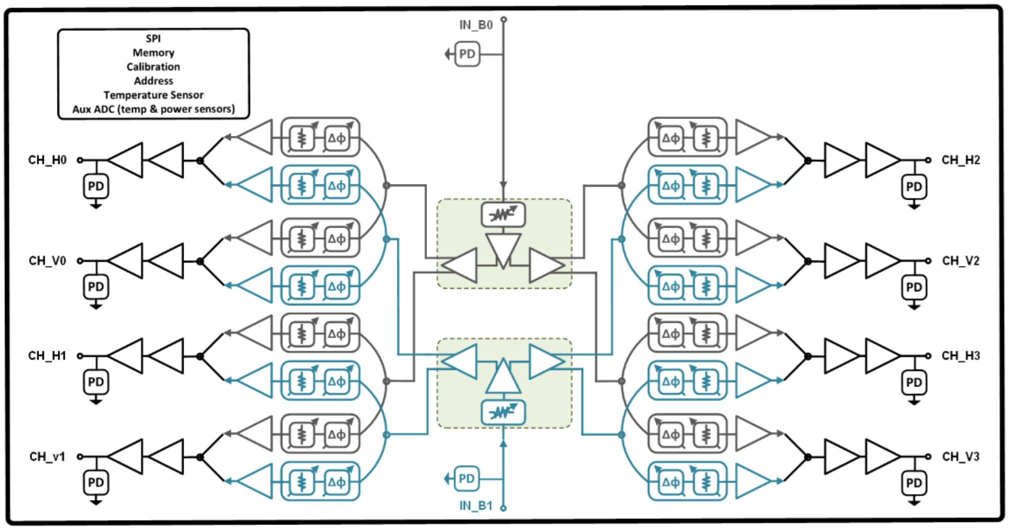


Figure 1: STAMPEDE Block Diagram

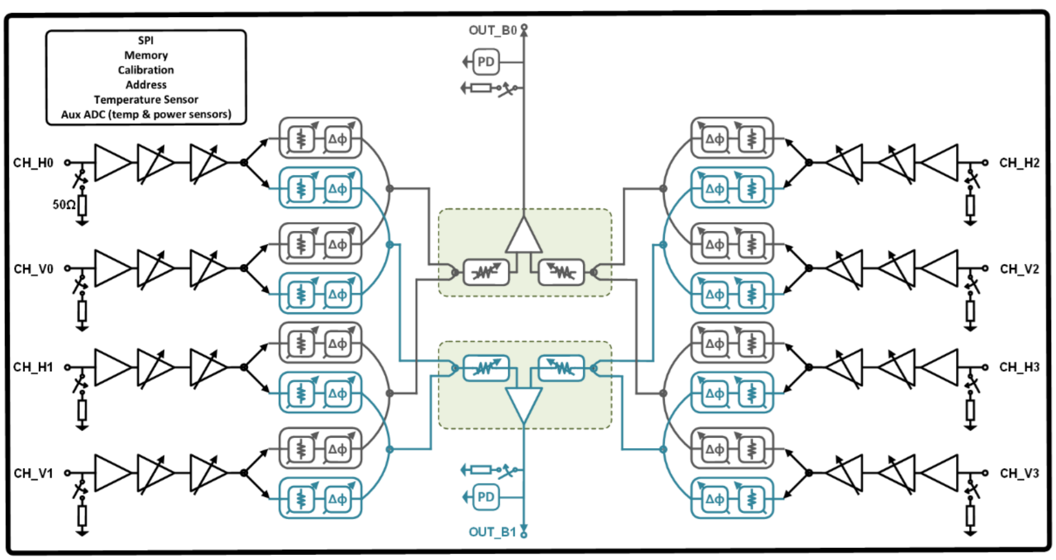


Figure 2: BLUEWAY Block Diagram

The channels are grouped into Quads which have common functionality, and the control registers are organized by Quad. Each Quad supports 2 channels (antenna interfaces) with two beam state per channel. The table below indicated the channel grouping:

Quad	Channels
1	H0: beam0, beam 1; V0: beam0, beam1
2	H1: beam0, beam 1; V1: beam0, beam1
3	H2: beam0, beam 1; V2: beam0, beam1
4	H3: beam0, beam 1; V3: beam0, beam1

The structure of a channel pair in a quad is shown below:

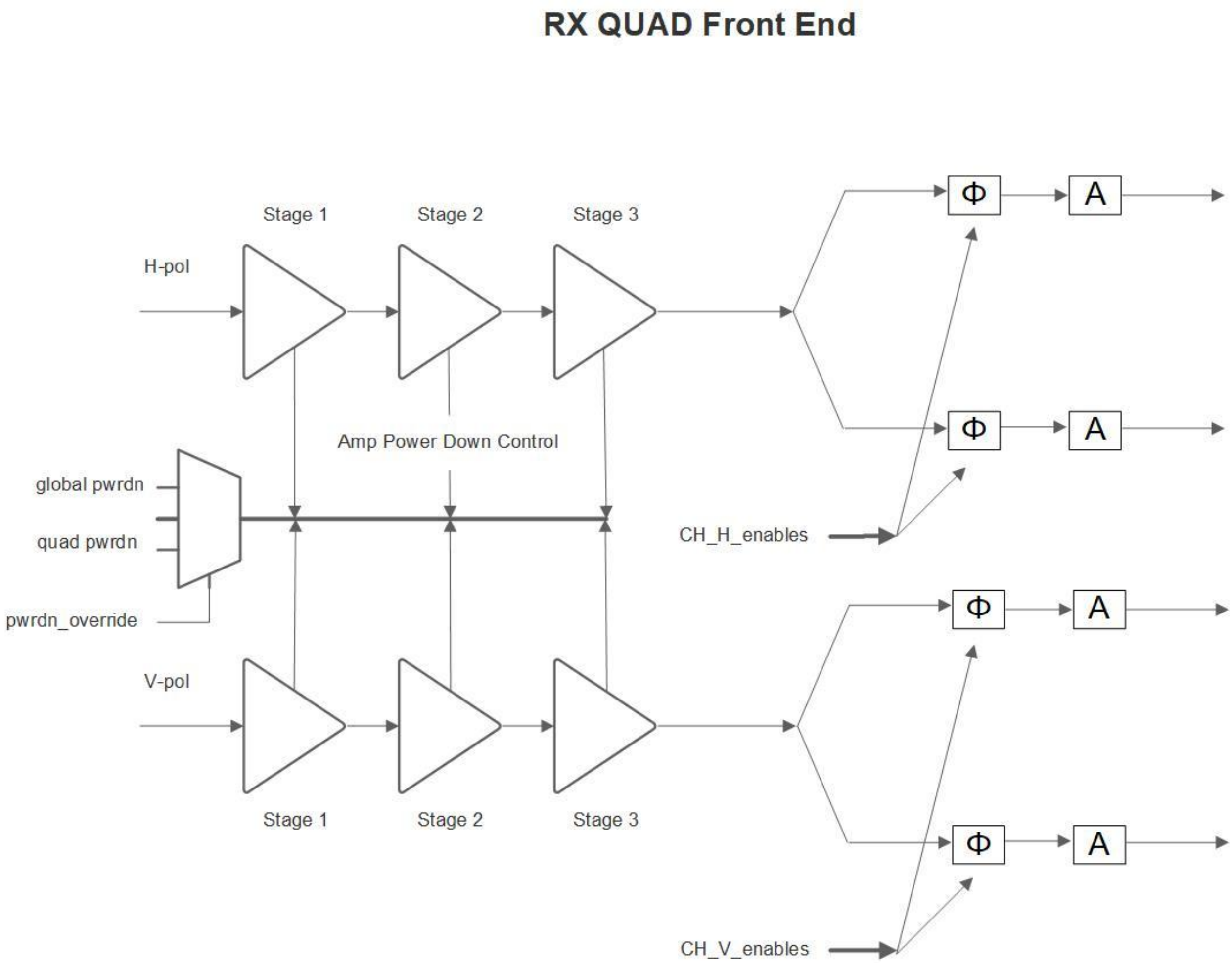


Figure 3: BLUEWAY QUAD block

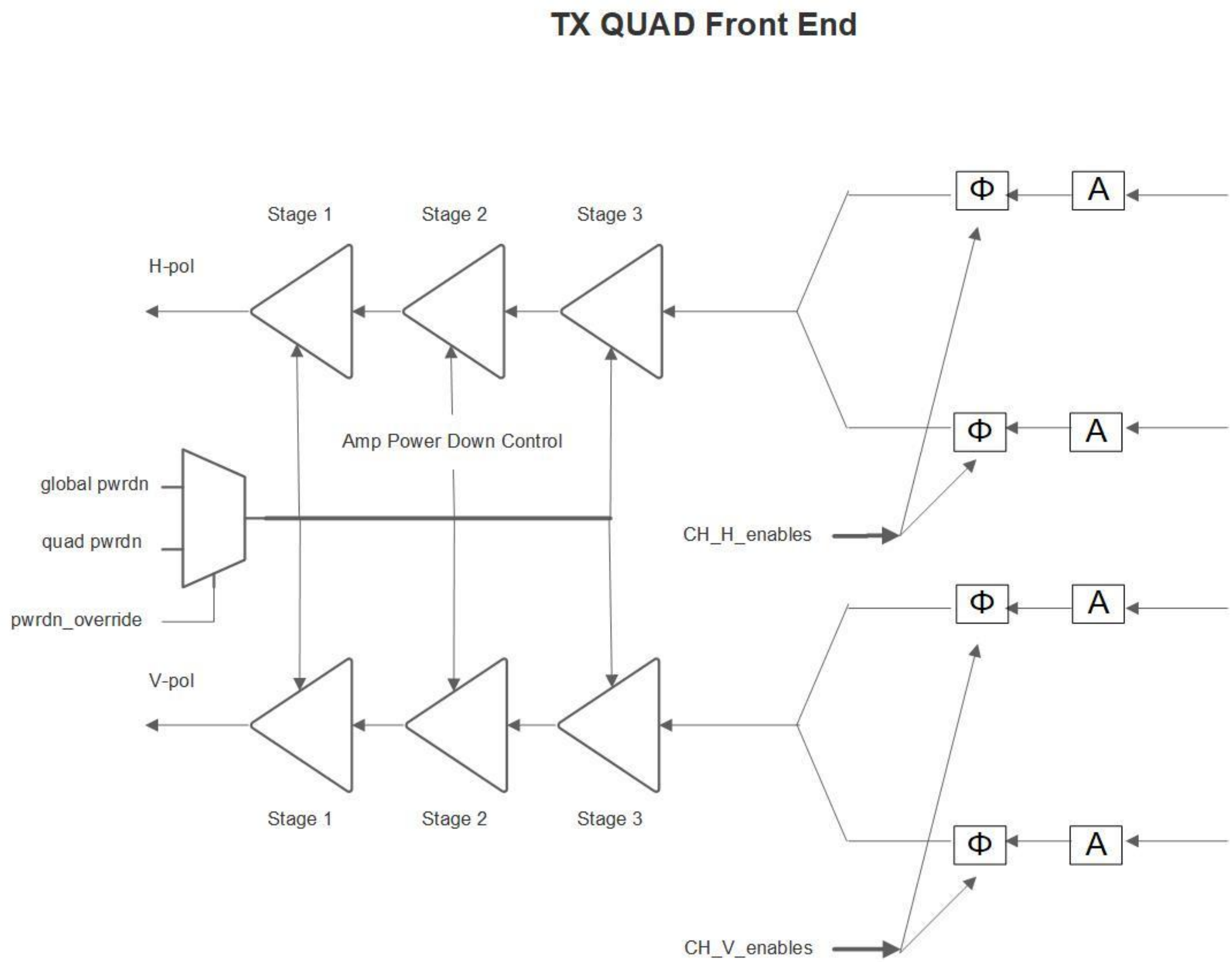


Figure 4: STAMPEDE QUAD Block

For each Quad there are enable bits, described below, which can be used to enable or disable the amps and switches for each channel. When the amps and switches are enabled (bit is set to 1) their ON or OFF state will be determined by the EN_STATE. And setting an enable bit to 0 will force the particular amp or switch OFF independent of the enable state (EN_STATE) disabling the function, and, in the case of the amplifiers, turning off the power. After reset, all amps and switches default to disabled.

Additionally, there are auto-power down operations for the BEAM and AMP enables. A particular BEAM will be disabled for both H and V if the beam attenuator is set to its maximum value of 127. If all the beams being associated with a PA or LNA are disabled, then that amplifier will be powered down as well. The logic for these operations is shown in the figure below:

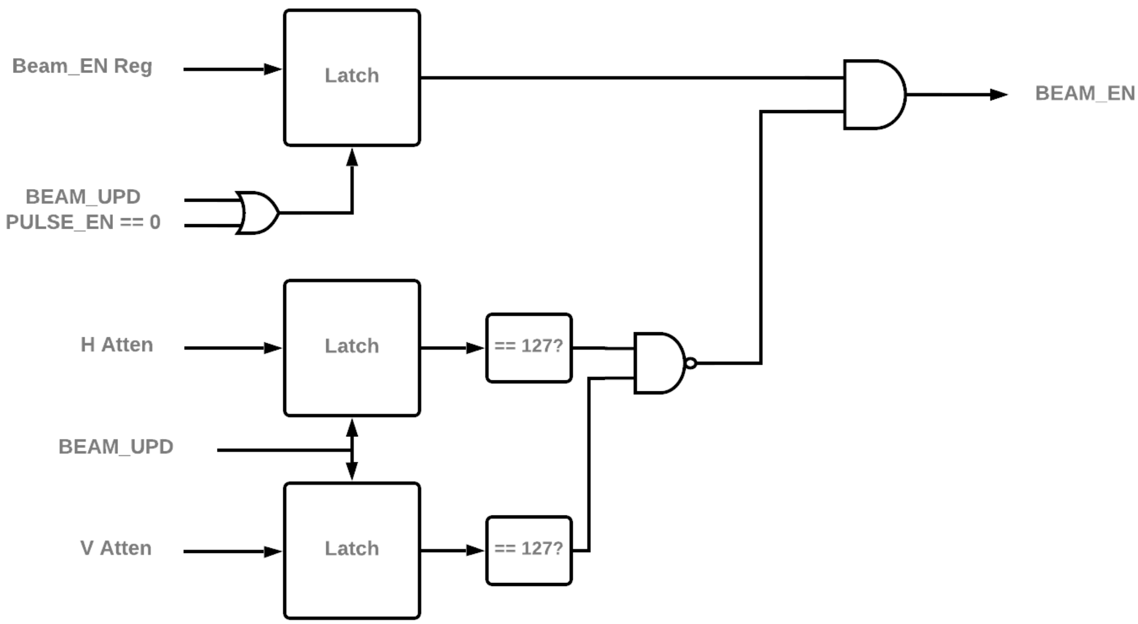


Figure 5: Beam auto disable.

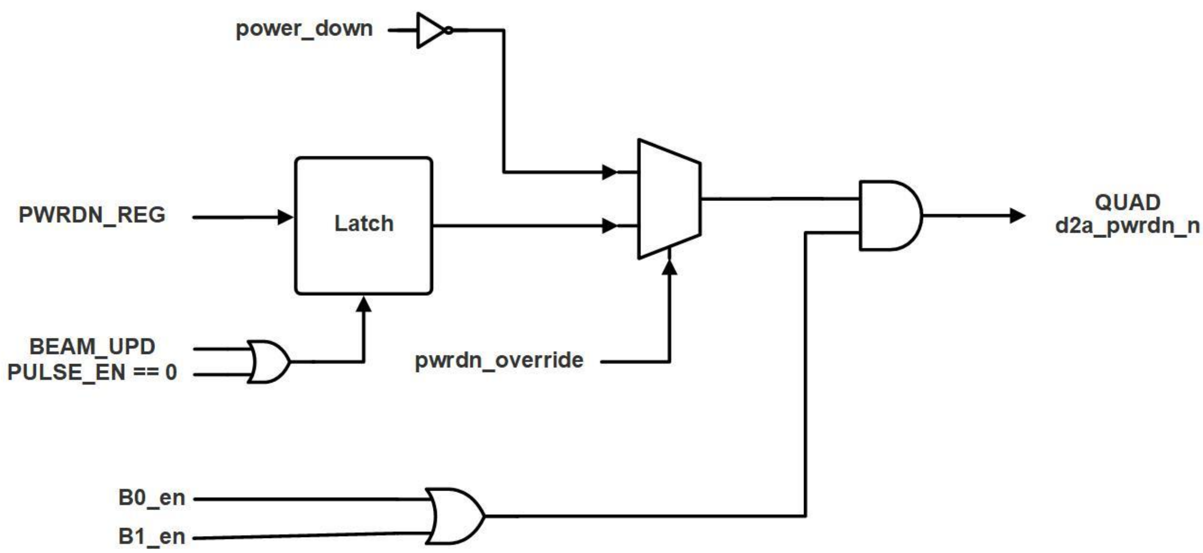


Figure 6: Amp auto powerdown

Each I/O port has a similar set of enables called Common enables, described below, which allow for the IO amps and switches to be disabled for each beam. These are disabled by default after reset. For a full beam path to operate at least 1 Quad beam and the Common amps and switches must be enabled.

Each channel can have its EN_STATE set independently. This can either by enabling global control of the EN_STATE by the Global_power_down bit or overriding the control through a programmable register, described below. In automatic mode, a value of 0 on the Global_power_down bit will configure the channels for OFF mode (i.e. turn OFF amps), and a value of 1 will configure the channels for ON mode (i.e. turn ON enabled amps). In override mode, the bits of the EN_OVERRIDE register determine which enabled amps are ON for each channel. The override operation is also set through for each channel through the EN_OVERRIDE register. After reset, the default operation of EN_STATE is override for all channels. The beam based auto power down will override either of these enable methods.

Each channel has multiple amplifier stages. Each amplifier stage has a bias current control which may be programmed through the SPI interface. The bias control is a 6-bit number with 0 being the minimum current and 63 being the maximum. There are also bias controls for the IO amplifiers and active combiners/splitters, which can be similarly set. After reset, the bias values are loaded from efuse to their default state which will be 0 before the efuse is programmed.

The beamforming phase and attenuation applied to a channel are selected from the large beam table stored on chip. This table can be thought of as a 4-channel x 1024-line memory with 16 bits of phase and attenuation control per channel (i.e. 64 bits x 1024 lines). A line pointer into this memory is used to select which set of beamforming coefficients are applied for each beam and channel. There are sixteen independent pointers which select beam0, beam1 for each of 8 channels (H and V). The pointers can select different lines within the memory or lines can be reused between beams and channels. Each pointer has a shadow register which is written by a SPI transaction and then applied to index the memory when the BEAM_UPDATE pin is pulsed high. If the BEAM_UPDATE pin is held at 1 then the writing of a pointer is applied immediately at the end of the SPI transaction. After reset, the beam pointers default to index 0.

Calibration of phase and gain is available for each beam on a per channel basis. There is a look up table for each channel which is loaded from efuse. Each look up table contains parameters for a calibration function to map from linear phase command to the optimal control word output for the phase shifter control.

Digital Interface Pinout

The figure below shows the top level of the SPI interface block. The pins are defined below

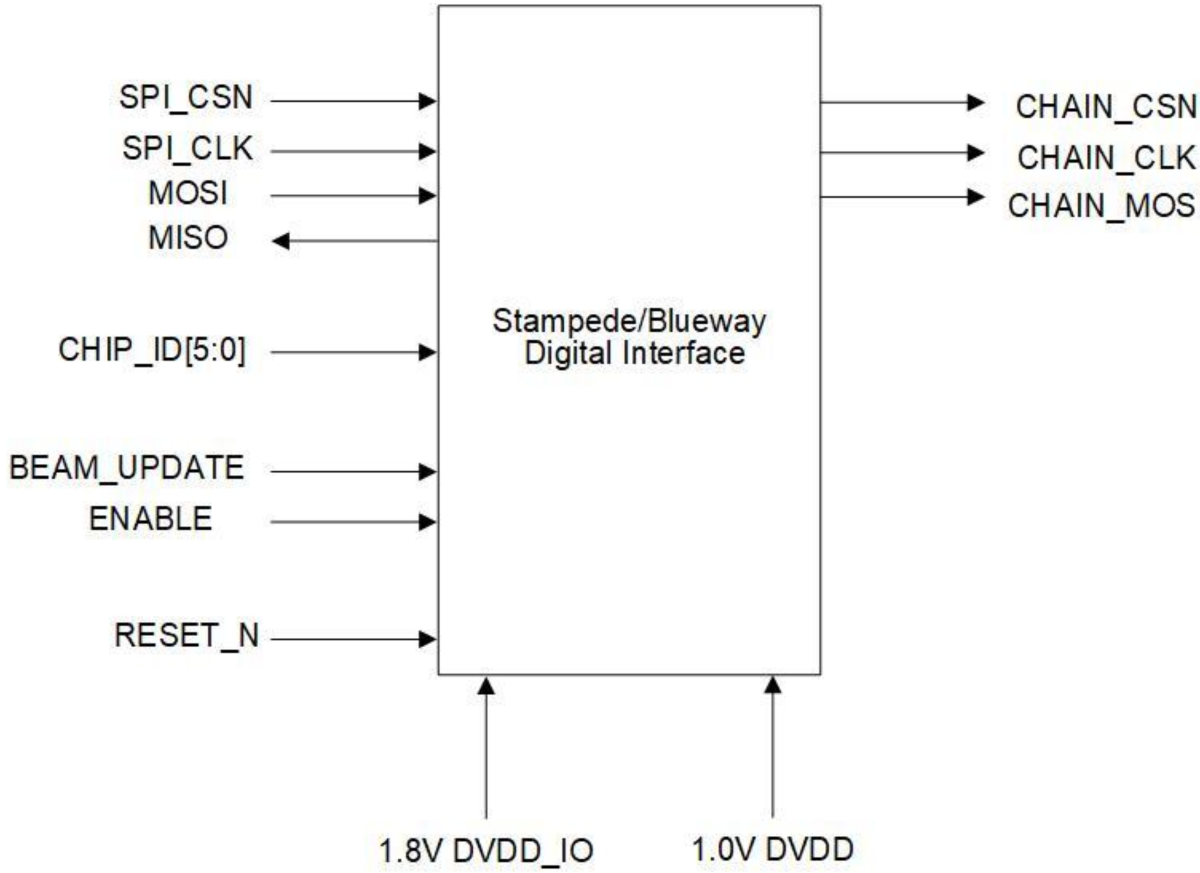


Figure 7: STAMPEDE/BLUEWAY Digital Interface

Pin Name	Direction	Connection	Function
SPI_CSN	In	external pad	SPI chip select, active low
SPI_CLK	In	external pad	SPI clock
SPI_MOSI	In	external pad	SPI data master out slave in
SPI_MISO	out	external tri-state	SPI data slave out master in
RESET_N	In	external pad	block reset, active low
BEAM_UPDATE	In	external pad	update beam index latch
CHAIN_CSN	out	external pad	Output buffer for SPI_CSN for daisy chain
CHAIN_CLK	out	external pad	Output buffer for SPI_CLK for daisy chain
CHAIN_DATA	out	external pad	Output buffer for SPI_MOSI for daisy chain
CHIP_ID[0]/ENUM_IN	In	external pad	CHIP_ID bit 0 multiplexed with Enum daisy chain input

CHIP_ID[1]/ENUM_OUT	In/Out	External pad	CHIP_ID bit 1 multiplexed with Enum daisy chain output
CHIP_ID[5:2]	In	external pad	CHIP_ID bits 2 to 5

Table 1: Digital Pinout

Array Interconnect

The STAMPEDE/BLUEWAY is designed to be used in an array with a single SPI interface addressing all the chips in the array. The intended interconnect is shown below in Figure 8 and Figure 9. In Figure 8 the SPI_CLK, SPI_CSN, SPI_MOSI, SPI_MISO, BEAM_UDPATE, and RESET_N are all wired in common to all the array devices. SPI_MISO is a tri-state buffer which is only active when a READ command is addressed to a particular device. This permits the devices to share the return bus.

Alternatively, as shown in Figure 9 the chips can be connected in a daisy chain configuration where the CHAIN outputs of one chip drive the inputs to the next chip and leads to simplified board routing. In this configuration the SPI_MISO is not used, instead during a read operation the output data is muxed onto the CHAIN_DATA output and propagates through the daisy chain. The CHAIN_DATA output of the last element in the chain should be connected to the master data input port.

The CHIP_ID lines are used to set individual address for the devices within the array and the SPI transaction allows for individual addressing to this address (Figure 8). Optionally the CHIP_ID can be enumerated by the devices through a daisy chain connection from CHIP_ID[1] of a device to CHIP_ID[0] of the next device in the chain. (Figure 9).

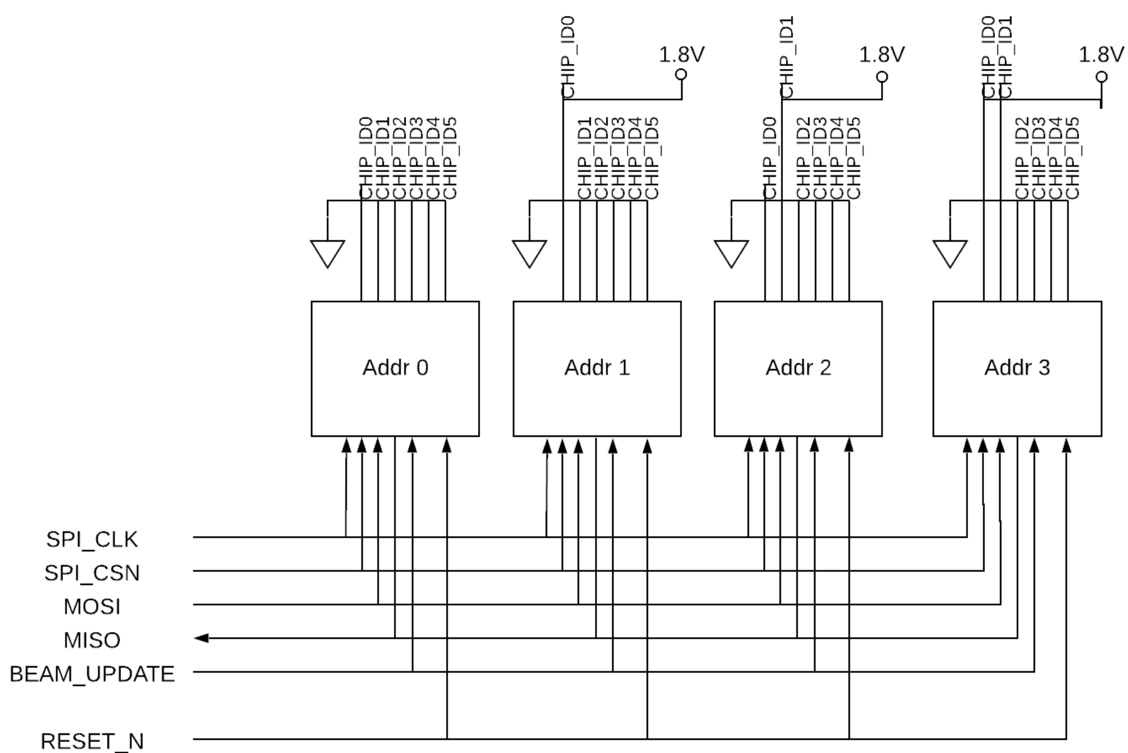


Figure 8: Array Interconnect

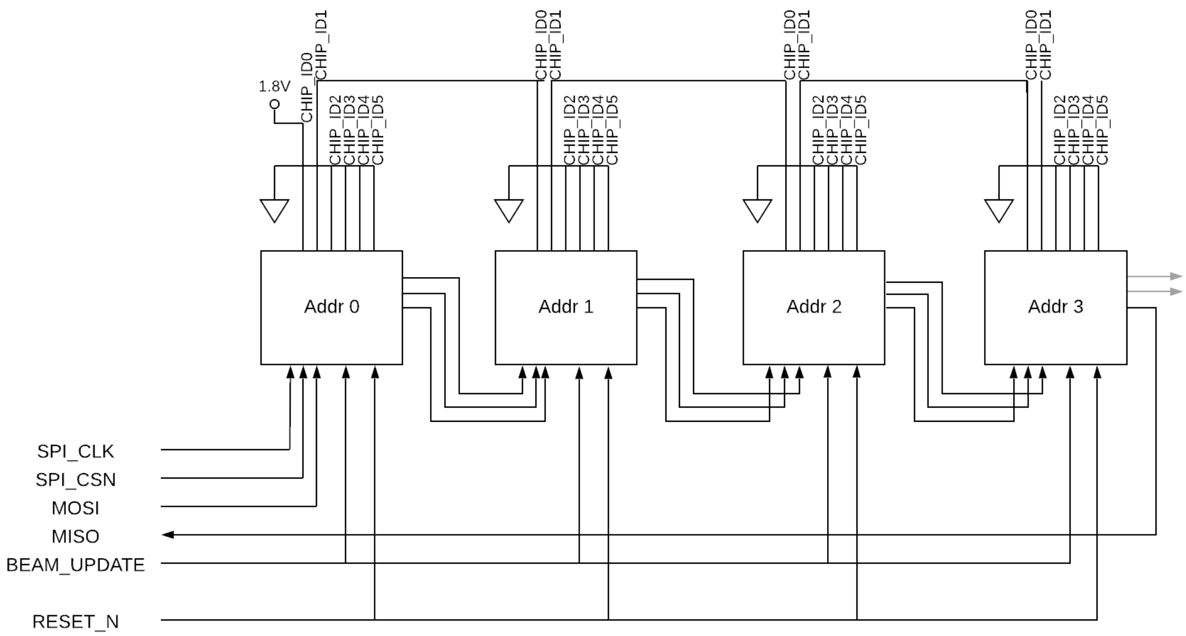


Figure 9: Array Interconnect with Enumeration

SPI Transaction

The SPI interface uses an atomic 32 or 56-bit transaction. The transaction is initiated by the chip-select line going low, followed by 32 or 56 clock rising edges, and terminated by the chip-select line rising. 56-bit transactions are only used with Unique ID addressing and are restricted to programming the CHIP ID. The general structure is shown in the figure below.

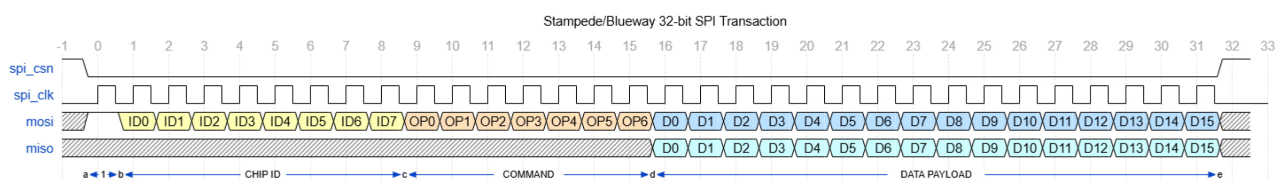


Figure 10: SPI transaction

The primary transaction uses a SHORT_ID (8-bits) to address a chip and is 32-bits long. There is also specialized command for programming the CHIP_ID that uses the device Unique ID for addressing and is 56-bit long. The contents of the SPI transaction are given in the table below. Data is sent in LSB first i.e. bit 0 corresponds to the first clock of the transaction.

Bits	Name	Description
0	ID Type	1 indicates 8 bits CHIP ID addressing
1-8	CHIP ID[0:7]	Chip identifier programmed by tying external pins high or low

9 – 15	Command[0:6]	See command table
16 – 31	Payload[0:15]	16-bit data payload

Table 2: 32-bit transaction

Bits	Name	Description
0	ID Type	0 indicates 32 bit Unique ID addressing
1-32	CHIP ID[0:31]	Chip identifier is 32-bit Unique ID
33 – 39	Command[0:6]	See command table
40 – 55	Payload[0:15]	16-bit data payload

Table 3: 56-bit transaction

The interface supports extended operations for read and write if the SPI_CSN remains low for additional clock cycles after the completion of the basic transaction. In the mode of operation the CHIP_ID and Command are retained from the start of the transaction and each subsequent set of 16 bits is treated as another payload. Thus for a write command, each subsequent payload would write into a register and cause the load pointer to auto increment.

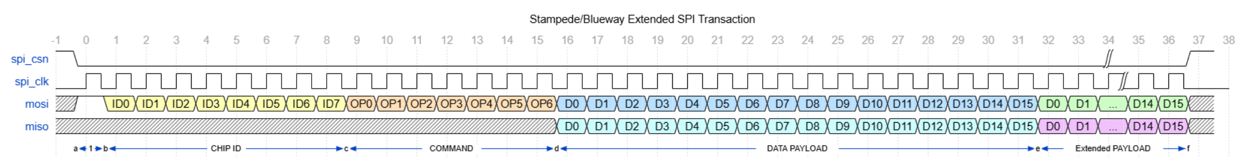


Figure 11: SPI extended transaction

SPI Address Space

The SPI registers are addressable 16 bits at a time and the address pointer indexes based on 16-bit words. The address space is contiguous across multiple subsections: beam table, static control bits, and ADC/ID. All registers are readable but writing to the ADC registers initiates an ADC update which is captured into the register.

Subsection	Address	Name	Description
Beam Table (read/write)	0x0000	Beam[0][0]	Beam Set 0 Phase and amplitude for Quad 0 [0:6]: attenuator setting [7:13]: phase shifter setting [15:14] Phase loss cal
	0x0001	Beam[0][1]	Beam Set 0 Phase and amplitude for Quad 1 [0:6]: attenuator setting [7:13]: phase shifter setting [15:14] Phase loss cal
	0x0002	Beam[0][2]	Beam Set 0 Phase and amplitude for Quad 2 [0:6]: attenuator setting [7:13]: phase shifter setting [15:14] Phase loss cal

	0x0003	Beam[0][3]	Beam Set 0 Phase and amplitude for Quad 3 [0:6]: attenuator setting [7:13]: phase shifter setting [15:14] Phase loss cal
	…	…	…
	0x0FFC	Beam[1023][0]	Beam Set 1023 Phase and amplitude for Quad 0 [0:6]: attenuator setting [7:13]: phase shifter setting [15:14] Phase loss cal
	0x0FFD	Beam[1023][1]	Beam Set 1023 Phase and amplitude for Quad 1 [0:6]: attenuator setting [7:13]: phase shifter setting [15:14] Phase loss cal
	0x0FFE	Beam[1023][2]	Beam Set 1023 Phase and amplitude for Quad 2 [0:6]: attenuator setting [7:13]: phase shifter setting [15:14] Phase loss cal
	0x0FFF	Beam[1023][3]	Beam Set 1023 Phase and amplitude for Quad 3 [0:6]: attenuator setting [7:13]: phase shifter setting [15:14] Phase loss cal
Version/Chip ID (read only)	0x1000	Version_ID[15:0]	Version identifier and Current CHIP setting either from external pins, enumeration, or written by unique ID Version[15:8] CHIP_ID[7:0]
Unique ID (read only)	0x1001	UID[15:0]	32-bit Unique ID for each chip
	0x1002	UID[16:31]	
Status Bits (read only)	0x1003	Status[4:0]	0: soft_reset in progress [1:4] reserved,
Output Daisy Amp Control (read/write)	0x1004		NA Bits [15:14,11:0] reserved bits Bits[13:12] = Beam[1:0] PD Supply power-down switches
Common Gain (read/write)	0x1005	B0_Common_gain[5:0] Pulse_en_1[6]	Common gain setting for each beam applied after combining (latch enabled with Pulse_en_1)
	0x1006	B1_Common_gain[5:0]	
Center Enables and power down	0x1008	Centerbias_en[0] Centermirror_en[1] Center_PD_override[2]	Enable for the center bias generator,

<table>
<tr><td></td><td></td><td>SPI_DAISY_CHAIN_DISABLE[3]
global_power_down[4]
Pulse_en_2[7]
Ds_dout[9 :8]
Ds_chain[11:10]
Center_EFUSE_PWRDN_DIS [12]ct</td><td>Override global power down for center,
Disable SPI daisy chain outputs,
Global power down for center and Quads,
(latch enabled with Pulse_en_2)
drive strength selection
EFUSE power down disable
(not latches)</td></tr>
<tr><td></td><td>0x1009</td><td>Beam0_DIST[7:0]
Center_bias_beam0_en[8]
Beam0_pwrdn[13:9]
Beam0_match[14]</td><td rowspan="2">Distribution Enables
Bias generation bit per beam
Power_down controls for override mode per beam
(latch enabled with Pulse_en_2)
See TABLE 5 below for Distribution Enables and Power_down Mapping specific for Blueway and Stampede</td></tr>
<tr><td></td><td>0x100A</td><td>Beam1_DIST[7:0]
Center_bias_beam1_en[8]
Beam1_pwrdn[13:9]
Beam1_match[14]</td></tr>
<tr><td></td><td>0x100B</td><td></td><td>Bit 8 is for Temp sensor ADC and DCTEST ADC Supply power-down switches</td></tr>
<tr><td rowspan="4">Quad Enables
(read/write)</td><td>0x100C</td><td>CH0_H_Enables[1:0]
CH0_V_Enables[9:8]
Pulse_en_CH0_1[11]</td><td rowspan="4">Amplifier and Switch Enables
H_beam0_en[0]
H_beam1_en[1]

V_beam0_en[0]
V_beam1_en[1]
(latch enabled with Pulse_en_CH*_1)</td></tr>
<tr><td>0x100D</td><td>CH1_H_Enables[1:0]
CH1_V_Enables[9:8]
Pulse_en_CH1_1[11]</td></tr>
<tr><td>0x100E</td><td>CH2_H_Enables[1:0]
CH2_V_Enables[9:8]
Pulse_en_CH2_1[11]</td></tr>
<tr><td>0x100F</td><td>CH3_H_Enables[1:0]
CH3_V_Enables[9:8]
Pulse_en_CH3_1[11]</td></tr>
<tr><td rowspan="3">Quad Power Down Control
(read/write)</td><td>0x1010</td><td>CH0_pwrdn[5:0]
CH0_pwrdn_override[6]
CH0_bias_EN[7]
Pulse_en_CH0_2[8]
CH0_EFUSE_PWRDN_DIS [9]</td><td rowspan="3">Power down override state for each channel
(latch enabled with Pulse_en_CH*_2)
CH*_EFUSE_PWRDN_DIS disables the auto power gating of effuse
(not latched)</td></tr>
<tr><td>0x1011</td><td>CH1_pwrdn[5:0]
CH1_pwrdn_override [6]
CH1_bias_EN[7]
Pulse_en_CH1_2[8]
CH0_EFUSE_PWRDN_DIS[9]</td></tr>
<tr><td>0x1012</td><td>CH2_pwrdn[5:0]
CH2_pwrdn_override [6]
CH2_bias_EN[7]
Pulse_en_CH2_2[8]</td></tr>
</table>

		CH0_EFUSE_PWRDN_DIS [9]	
	0x1013	CH3_pwrdn[5:0] CH3_pwrdn_override [6] CH3_bias_EN[7] Pulse_en_CH3_2[8] CH0_EFUSE_PWRDN_DIS [9]	
Calibration frequency adjustment	0x1014	CH0_Cal_freq_sel_beam0[2:0] CH0_Cal_freq_sel_beam1[5:3] Pulse_en_CH0_3[9]	Frequency selection for calibration optimization (latch enabled with Pulse_en_CH*_3)
	0x1015	CH1_Cal_freq_sel_beam0[2:0] CH1_Cal_freq_sel_beam1[5:3] Pulse_en_CH1_3[9]	
	0x1016	CH2_Cal_freq_sel_beam0[2:0] CH2_Cal_freq_sel_beam1[5:3] Pulse_en_CH2_3[9]	
	0x1017	CH3_Cal_freq_sel_beam0[2:0] CH3_Cal_freq_sel_beam1[5:3] Pulse_en_CH3_3[9]	
Gain Control per Channel (read/write)	0x1018	Gain_control_H0[3:0] Gain_control_V0[11:8] Pulse_en_CH0_4[12]	4 bit gain control for two channels per quad (latch enabled with Pulse_en_CH*_4)
	0x1019	Gain_control_H1[3:0] Gain_control_V1[11:8] Pulse_en_CH1_4[12]	
	0x101A	Gain_control_H2[3:0] Gain_control_V2[11:8] Pulse_en_CH2_4[12]	
	0x101B	Gain_control_H3[3:0] Gain_control_V3[11:8] Pulse_en_CH3_4[12]	
Power Detector ADC Enables (read/write)	0x101C	PD control Quad0[3:0] Dctest_EN Quad0[7:4]	ADC enables for Quads H_PD_EN[0] H_ENBGR[1] V_PD_EN[2] V_ENBGR[3] Dctest_EN[7:4]
	0x101D	PD control Quad1[3:0] Dctest_EN Quad1[7:4]	
	0x101E	PD control Quad2[3:0] Dctest_EN Quad2[7:4]	
	0x101F	PD control Quad3[3:0] Dctest_EN Quad3[7:4]	
ADC CLK/MASK (read/write) IO PD enables	0x1020	ADC_DIV[3:0] ADC_CLK_POL	4 – bit clock divider selection, 2^N 0000 = 1 0001 = 2 (recommended value) 0010 = 4 0011 = 8 0100 = 16 0101 = 32 0110 = 64 0111 = 128

			1000 = 256
	0x1021	ADC_MASK[13:0] ADC_RESET_POL ADC_AUTO_RESET_DISABLE	[13:0] 8 bit mask for ADCs to start [14] ADC reset polarity 0 = active low, 1 = active high
Temp Cal (read/write)	0x1022	Temp Sensor Cal/Contol	Slope and offset calibration for Temp Sensor [0:3] temp sensor offset cal [4:7] temp sensor slope cal [8] temp sensor core enable [9] temp sensor band gap enable
ADC Registers (read only)	0x1023	Temp_ADC[7:0]	8-bit value from ADC 13 (temp sensor)
	0x1024	PD_ADC_H0[7:0]	8-bit value from ADC 0 (PD H0)
	…		
	0x1027	PD_ADC_H3[7:0]	8-bit value from ADC 3 (PD H3)
	0x1028	PD_ADC_V0[7:0]	8-bit value from ADC 4 (PD V0)
	…		
	0x102B	PD_ADC_V3[7:0]	8-bit value from ADC 7 (PD V3)
	0x102C	PD_ADC_PORT0[7:0]	8-bit value from ADC 8 (PD PORT0)
	0x102D	PD_ADC_PORT1[7:0]	8-bit value from ADC 9 (PD PORT1)
	0x1030	TEST_ADC[7:0]	8-bit value from test adc
	0x1031	Dctest_EN center [9:0] Dctest_ENBGR[10] Dctest_GC[12:11] Dctest_mode[15 :13]	Select test ADC source
	0x1032	PD_EN_Ports[7:0]	Power detector enables for port detectors
	0x1033	PD GC Ports[7:0]	Power detector GC for Port detectors
Power Detector GC Setting (read/write)	0x1034	PD GC Quad0[0:3]	[0:1] PD H0 gain [2:3] PD V0 gain
	0x1035	PD GC Quad1[0:3]	[0:1] PD H1 gain [2:3] PD V1 gain
	0x1036	PD GC Quad2[0:3]	[0:1] PD H2 gain [2:3] PD V2 gain
	0x1037	PD GC Quad3[0:3]	[0:1] PD H0 gain [2:3] PD V0 gain
Bias Control Registers (read/write)	0x1038		Amplifier bias setting 2 – 6 bit settings per address, in bits [0:5] and [8:13]. See Table 9 below for complete amp listing
	…	…	
	0x104B		
Bias Control Beams (read/write)	0x104C	B0_bias	Amplifier bias for beam amplifier [0:5] stage 1 [8:13] stage 2_0 [0:5] stage2_1 [8:6] stage1_cbias
	0x104D		
	0x104E	B1_bias	
	0x104F		

			[11:9] stage2_0_cbias [14:12]stage2_1_cbias
Global Bias Control Registers (read/write)	0x1050	Global_Bias	Bits [3:0] ptat_slope1[3:0] Bits[11:8] ptat_slope2[3:0]
	0x1051	Global_Bias	Bits[3:0] ptat_slope3[3:0]
Dist Bias (read/write)	0x1052	DIST_BIAS[5:0]	Bias for distribution amplifier
	0x1053	reserved	
Daisy Chain Amp Bias (read/write)	0x1054	Daisy_chain0_amp_bias[5:0] Daisy_chain1_amp_bias[13:8]	Daisy chain bias and tuning
	0x1055	Daisy_chain2_amp_bias[5:0] Daisy_chain3_amp_bias[13:8]	
	0x1056	Daisy_chain0_tuning[3:0] Daisy_chain0_offsetbias[4] Daisy_chain0_cbias[7:5] Daisy_chain1_tuning[11:8] Daisy_chain1_offsetbias[12] Daisy_chain1_cbias[15:13]	
	0x1057	Daisy_chain2_tuning[3:0] Daisy_chain2_offsetbias[4] Daisy_chain2_cbias[7:5] Daisy_chain3_tuning[11:8] Daisy_chain3_offsetbias[12] Daisy_chain3_cbias[15:13]	
Extra (read/write)	0x1058	d2a_extra_bits_B0[7:0] d2a_extra_bits_B1[15:8]	Extra bits from center block
	0x1059	Ptat bias and slope	Bits [4:0] ptat_bias_cal[4:0] Bit [5] ptat_slope_x2_en
	0x105A	d2a_extra_bits_misc[15:0]	Extra bits from center block
	0x105B	reserved	
Beam Calibration (read/write)	0x105C	H_b0_atten_cal[3:0], H_b1_atten_cal[11:8], V_b0_atten_cal[3:0], V_b1_atten_cal[11:8]	Calibration bits for the beam table outputs to RTPS, repeated for each Quad. Detailed mapping is given below
	…		
	0x1067		
Cap Tuning (read/write)	0x1068	Q1_captune[15:0] … Q4_captune[15:0]	12 bits of tuning for each Quad
	…		
	0x106B		

Phase Calibration (read/write)	0x106C	Calibration function parameters	Phase calibration look up table
	…		
	0x1183		
ADC Debug	0x1184	ADC_CLK/MASK_Q0, ADC_CLK/MASK_Q1, ADC_CLK/MASK_Q2, ADC_CLK/MASK_Q3	Quad debug readback of ADC registers
	…		
	0x118B		
Quad Beam Index Debug (read only)	0x118C	H beam 0 index Quad1	Readback of beam table index pointers local to each Quad
	0x118D	H beam 0 index Quad2	
	0x118E	H beam 0 index Quad3	
	0x118F	H beam 0 index Quad4	
	0x1190	H beam 1 index Quad1	
	0x1191	H beam 1 index Quad2	
	0x1192	H beam 1 index Quad3	
	0x1193	H beam 1 index Quad4	
	0x1198	V beam 0 index Quad1	
	0x1199	V beam 0 index Quad2	
	0x119A	V beam 0 index Quad3	
	0x119B	V beam 0 index Quad4	
	0x119C	V beam 1 index Quad1	
	0x119D	V beam 1 index Quad2	
	0x119E	V beam 1 index Quad3	
	0x119F	V beam 1 index Quad4	
	0x11A4	H beam 0 set Quad1	Readback phase and atten outputs [6:0] attenuator setting [15:7] phase setting
	0x11A5	H beam 0 set Quad2	
	0x11A6	H beam 0 set Quad3	
	0x11A7	H beam 0 set Quad4	
	0x11A8	V beam 0 set Quad1	
	0x11A9	V beam 0 set Quad2	
	0x11AA	V beam 0 set Quad3	
	0x11AB	V beam 0 set Quad4	
	0x11AC	H beam 1 set Quad1	
	0x11AD	H beam 1 set Quad2	
	0x11AE	H beam 1 set Quad3	
	0x11AF	H beam 1 set Quad4	
	0x11B0	V beam 1 set Quad1	
	0x11B1	V beam 1 set Quad2	
	0x11B2	V beam 1 set Quad3	
	0x11B3	V beam 1 set Quad4	
Efuse Quad Debug	0x11BC		
	…		
	0x11DB		
Efuse raw data (read only)	0x11DC	Efuse_data[0:15]	512 bits read from efuse divided into 32 16-bit words
	…	…	
	0x11FB	Efuse_data[496:511]	
	0x11FC	Efuse_row[0:3]	Efuse row pointer

Efuse control registers (read only)	0x11FD	Efuse_prog_cnt[0:10]	Efuse programming timer, number of SPI_CLK cycles for programming, nominal value = 1000
	0x11FE	Efuse_f_val[0:5]	Resistance value for efuse sense, nominal values 00010
	0x11FF	Efuse_status[0:7]	0: efuse data valid, 1: efuse_sense_busy, 2: efuse_prog_busy, 3: efuse_gatec, 4: efuse_mimicn, 5:7: efuse_time
	0x1200	Efuse_prog_data[0:15]	32 bits of column data to be programmed into row
	0x1201	Efuse_prog_data[16:31]	
	0x1202	Efuse_sense[0:15]	32 bits of data from most recent Efuse sense operation
	0x1203	Efuse_sense[16:31]	

Table 4: Address Space

Bit Names	Size	RX	TX	Blueway Function	Stampede Function
DIST_B0_St1_en	4	yes	yes	CH enable signals for the 2-to-1 DIST_St1 at Beam0; 0x1009[0] for the signal path of CH1-0 from the left half array, 0x1009[1] for the signal path of CH3-2 from the right half array, 0x1009[3:2] for reserved bits;	1 for enable and 0 for disable; 0x1009[0] for CH0, 0x1009[1] for CH1, 0x1009[2] for CH2, 0x1009[3] for CH3 for the 1-to-4 distribution network at Beam 0;
DIST_B1_St1_en	4	yes	yes	CH enable signals for the 2-to-1 DIST_St1 at Beam1; 0x100A[0] for the signal path of CH1-0 from the left half array, 0x100A[1] for the signal path of CH3-2 from the right half array, 0x100A[3:2] for reserved bits;	1 for enable and 0 for disable; 0x100A[0] for CH0, 0x100A[1] for CH1, 0x100A[2] for CH2, 0x100A[3] for CH3 for the 1-to-4 distribution network at Beam 1;
DIST_B0_St2_0_en	2	yes	no	0x1009[5:4] reserved bits due to passive Wilkinson combiner;	0x1009[5:4] reserved bits
DIST_B1_St2_0_en	2	yes	no	0x100A[5:4] reserved bits due to passive Wilkinson combiner;	0x100A[5:4] reserved bits
DIST_B0_St2_1_en	2	yes	no	0x1009[7:6] reserved bits due to passive Wilkinson combiner;	0x1009[7:6] reserved bits
DIST_B1_St2_1_en	2	yes	no	0x100A[7:6] reserved bits due to passive Wilkinson combiner;	0x100A[7:6] reserved bits
Beam0_pwrdn	5	yes	yes	0x1009[11] for DIST_St1 at Beam0; 0x1009[13:12,10:9] reserved bits;	0x1009[9] for distribution network at Beam 0; 0x1009[13:10] reserved bits;

Beam1_pwrdn	5	yes	yes	0x100A[11] for DIST_St1 at Beam1; 0x100A[13:12,10:9] reserved bits;	0x100A[9] for distribution network at Beam 1; 0x100A[13:10] reserved bits;

Table 5: STAMPEDE/BLUEWAY Enable and Power Down assignment

SPI Commands

The interface uses the 7 bit command word to determine how the payload data is used. The commands are summarized in the table below.

Command	Value	Description
LOAD_POINTER	7'b0000000	Load read/write address pointer from payload
READ_REGISTER	7'b0000001	Read from address pointer location
WRITE_REGISTER	7'b0000010	Write to address pointer location
WRITE_H0_BEAM0_IDX	7'b0100000	Write beam set pointer for beam0 H0
WRITE_V0_BEAM0_IDX	7'b0100001	Write beam set pointer for beam0 V0
WRITE_H1_BEAM0_IDX	7'b0100010	Write beam set pointer for beam0 H1
WRITE_V1_BEAM0_IDX	7'b0100011	Write beam set pointer for beam0 V1
WRITE_H2_BEAM0_IDX	7'b0100100	Write beam set pointer for beam0 H2
WRITE_V2_BEAM0_IDX	7'b0100101	Write beam set pointer for beam0 V2
WRITE_H3_BEAM0_IDX	7'b0100110	Write beam set pointer for beam0 H3
WRITE_V3_BEAM0_IDX	7'b0100111	Write beam set pointer for beam0 V3
WRITE_H0_BEAM1_IDX	7'b0101000	Write beam set pointer for beam1 H0
WRITE_V0_BEAM1_IDX	7'b0101001	Write beam set pointer for beam1 V0
WRITE_H1_BEAM1_IDX	7'b0101010	Write beam set pointer for beam1 H1
WRITE_V1_BEAM1_IDX	7'b0101011	Write beam set pointer for beam1 V1
WRITE_H2_BEAM1_IDX	7'b0101100	Write beam set pointer for beam1 H2
WRITE_V2_BEAM1_IDX	7'b0101101	Write beam set pointer for beam1 V2
WRITE_H3_BEAM1_IDX	7'b0101110	Write beam set pointer for beam1 H3
WRITE_V3_BEAM1_IDX	7'b0101111	Write beam set pointer for beam1 V3
ADC_CAPTURE	7'b0000100	Initiate ADC sampling
INIT_ENUM	7'b0000101	Initiate ID enumeration
SOFT_RESET_CMD	7'b0000111	Reset all writable registers to default values
WRITE_SHORT_ID	7'b0001000	Write short CHIP ID
EFZ_ROW_SET	7'b1000000	Set row pointer for efuse sensing or programming
EFZ_SET_DATA_LOW	7'b1000001	Set low 16 bits of efuse column data for programming
EFZ_SET_DATA_HIGH	7'b1000010	Set high 16 bits of efuse column data for programming
EFZ_F_VAL	7'b1000100	Set the F-value for sense resistance
EFZ_PROG_TIME	7'b1000110	Set the programming timer value
EFZ_SET_CHAR	7'b1000111	Set the state of the characterization control bits GATEC, MIMICN, TIME

EFZ_PROG_ROW	7'b1100000	Program the current row with column data
EFZ_SENSE_ROW	7'b1001000	Sense the current row
EFZ_SENSE_ALL	7'b1001001	Read the entire contents of efuse

Table 6: SPI Commands

Read and write access to the address space is by indirect addressing through an address pointer register. To access a specific location this register must be programmed prior to the **READ_REGISTER**/**WRITE_REGISTER** command via the **LOAD_POINTER** command. The value stored in the pointer will auto increment with each **READ_REGISTER**/**WRITE_REGISTER** command to facilitate sequential access without the need for additional **LOAD_POINTER** commands. This pointer will wrap around to 0 after the last address is accessed (0x11FF) and is initialized to 0 by reset.

Loading the address pointer (**LOAD_POINTER** command = 0x00) will load the 13 LSBs of the payload into the read/write set register. No other registers are updated.

Loading the one of the beam set registers (**WRITE_*_BEAM_IDX** command = 0x20 – 0x37) will load the 10 LSBs of the payload into a shadow register to select the beam coefficient set output to the beam former. The actual output is not updated until the BEAM_UPDATE signal is pulsed as described below. There are twenty-four beam set registers which are loaded independently.

Writing a register (**WRITE_REGISTER** command = 0x02) will load the 16-bit payload into the address pointed to by the address pointer. The address pointer is incremented by one and will wrap around to 0 if it is currently at the highest address (0x11FF). The writing of the register and the increment of the pointer occur on the falling edge of the last clock of the transaction. Note many registers are not a full 16-bit value and the extra bits are not implemented.

Reading a register (**READ_REGISTER** command = 0x01) will load the register currently pointed to by the address pointer into an output shift register, whose bits are shifted out on SPI_DOUT coincident with the payload bits during a transaction. The address pointer is incremented by one and will wrap around to 0 if it is currently at the highest address. SPI_MISO will be held in a high-Z state until the active payload portion of the read command.

The temperature sensor and power detector ADC updates will be initiated by the **ADC_CAPTURE** command (command = 0x08). The process will start on the 16^{th} clock of the transaction with the sampling of data, and the ADC values will be updated 25*2^ADC_CLK_DIV cycles later. The SPI_CLK should continue to be active during the ADC capture process although the SPI_CSN does not need to remain active. The payload is not used for this command.

The **INIT_ENUM** command (command = 0x09) places the device into a state in which it will perform an enumeration operation on the next SPI_CSN transaction. The enumeration process is described in more detail below. The payload value is not used by this command.

SOFT_RESET (command = 0x0F) will reset all the registers to their default values. The payload is not used by this command.

WRITE_SHORT_ID (command = 0x10) uses unique ID addressing (56-bit transaction) to overwrite the CHIP_ID value. The 8 LSBs of the payload are loaded into the device CHIP_ID at the completion of this command.

The **EFZ_ROW_SET** command (command = 0x40) sets the row pointer for efuse sensing or programming. This is a 4-bit number loaded from the 4 LSBs of the payload.

The **EFZ_SET_DATA_LOW** command (command = 0x41) will set the lower 16 bits of the efuse column programming register from the payload. During programming, the column locations with a 1 in current row will be blown.

The **EFZ_SET_DATA_HIGH** command (command = 0x42) will set the upper 16 bits of the efuse column programming register from the payload. During programming, the column locations with a 1 in current row will be blown.

The **EFZ_F_VAL** command (command = 0x44) will load the 6-bit F-value register for the efuse from the LSBs of the payload. The F-value controls the resistance threshold of the sense amps during efuse readback.

The **EFZ_PROG_TIME** command (command = 0x46) will load the 11 LSBs of the payload into a programming timer. The timer determines how long the PROGP signal is applied for each column during a programing operation.

The **EFZ_SET_CHAR** command (command = 0x47) sets signal used for characterization testing of the efuse IP. The bits are load from the 5 LSBs as mapped below

GATEC	Payload[0]
MIMICN	Payload[1]
TIME[0:2]	Payload[4:2]

Table 7: Characterization Bits

The **EFZ_PROG_ROW** command (command = 0x60) initiates the efuse programming sequence. This is described in further detail below. The SPI_CLK must remain active for the entire programming sequence although the SPI_CSN does not need to remain active after the transaction has initiated programming.

The **EFZ_SENSE_ROW** command (command 0x48) will read the efuse content for the row pointed to by the current row pointer. The SPI_CLK must remain active during the entire sense process although the SPI_CSN does not need to remain active after the transaction has initiated sensing.

The **EFZ_SENSE_ALL** command (command 0x49) will read the entire contents of the efuse cycling through rows 0 to 15. The SPI_CLK must remain active during the entire sense process although the SPI_CSN does not need to remain active after the transaction has initiated sensing.

SPI Special Functions

The SPI interface has additional control lines and internal control functions to implement special features of the interface.

RESET_N: is an active low reset signal which puts the SPI interface into a known state. The address pointer, and both beam set registers are all set to 0. The CNTL bits are initialized to a default state as are the bits of Beam Set 0. All other beam set bits are defaulted to 0. Reset will also initiate an

EFZ_SENSE_ALL and the SPI_CLK must be active after RESET_N is released to allow the EFUSE to be read and the calibration bits initialized.

BEAM_UPDATE: this line will load the beam coefficient set into a latch which is used to select the Beam Set output to the analog circuits when it is pulsed high. This line controls a transparent latch so if it is held high then a write to a beam set register will take effect immediately. This line allows for precise control of the beam switching time. If this line is held high the beam former parameters will be updated when a beam set write is completed. This line can also be used to control the update of certain enables, gains controls, and power down bits. These registers will not be updated unless BEAM_UPDATE is high when the associated PULSE_EN bit is set in the control register.

ADC sampling: the ADCs associated with the power detectors and the temperature detectors will only execute a conversion in response to ADC_CAPTURE command. The ADC value will be updated into the ADC register at the end of the transaction and can be read with a subsequent transaction.

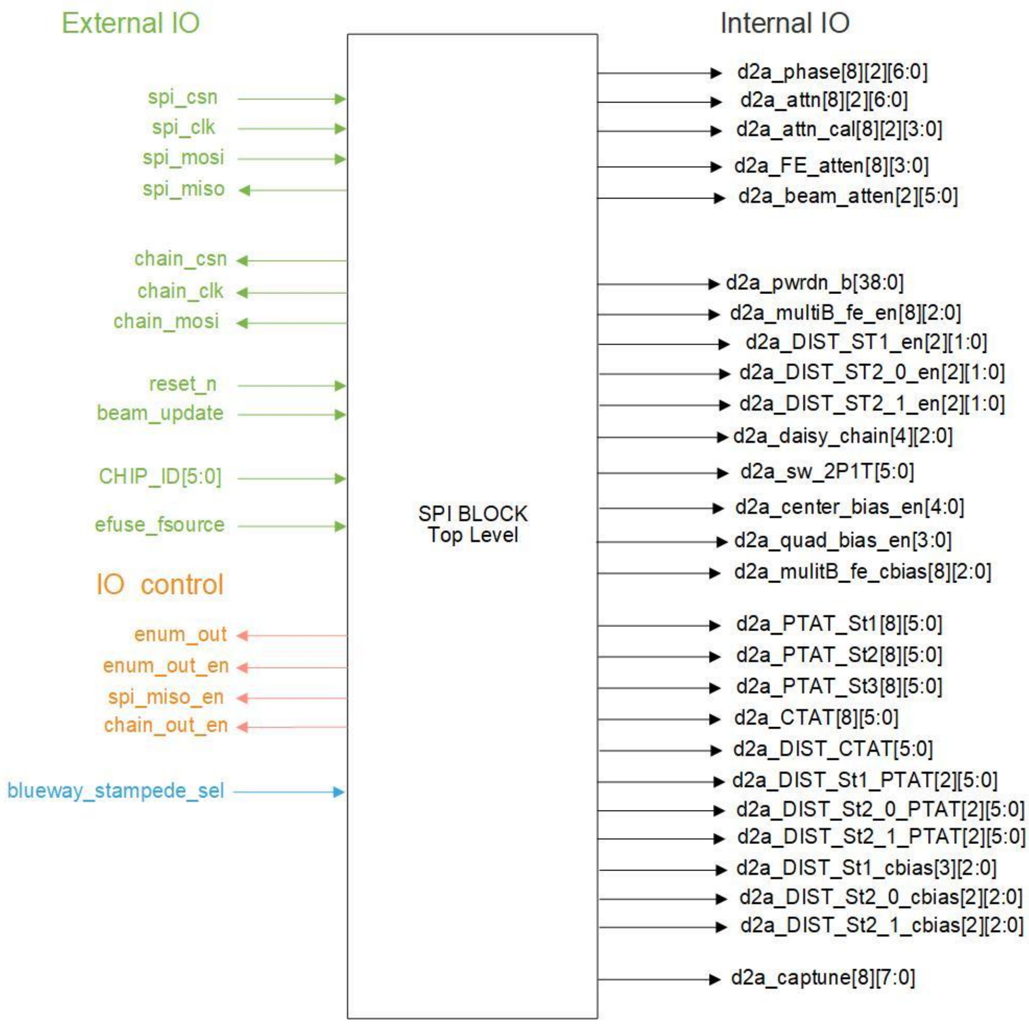
CHIP_ID: there are 6 external pins which are used to set the 6 LSBs of the SHORT ID for any device. Upon reset the device SHORT ID will be set based on these pins with the 2 MSBs set to 0. This ID can be overwritten by using the devices Unique ID with a WRITE_SHORT_ID command. Additionally, the device enumeration process can be used to change the SHORT_ID. A SPI transaction will only be executed if the 8 ID bits of the transaction match the SHORT ID setting.

BROADCAST ID: the SHORT ID of “11111111” is a broadcast id and all chips will respond to a transaction beginning with this value. This is intended to update the beam set or enable register for all chips in an array without addressing them individually, but it also applies to other register writes. Reads are not supported from the broadcast ID to avoid drive conflict of multiple chips in a wired-or connection of SPI_MISO.

Calibration and Efuse: The SPI interface includes many bits for calibration and static control, such as bias settings for the many amplifiers. These bits are addressable via SPI read and write transactions. They can also be initialized from an embedded efuse block. It is intended that the optimal values be programmed into the efuse during manufacturing test and the loaded into the calibration bits upon reset. The SPI interface also includes a mechanism for programming and reading back the efuse data.

SPI Block Top Level

The figure below shows the top level of the SPI interface block. The pins are defined below.


External IO
Internal IO
spi_csn
spi_clk
spi_mosi
spi_miso
chain_csn
chain_clk
chain_mosi
reset_n
beam_update
CHIP_ID[5:0]
efuse_fsource
IO control
enum_out
enum_out_en
spi_miso_en
chain_out_en
blueway_stampede_sel
SPI BLOCK
Top Level
d2a_phase[8][2][6:0]
d2a_attn[8][2][6:0]
d2a_attn_cal[8][2][3:0]
d2a_FE_atten[8][3:0]
d2a_beam_atten[2][5:0]
d2a_pwrdn_b[38:0]
d2a_multiB_fe_en[8][2:0]
d2a_DIST_ST1_en[2][1:0]
d2a_DIST_ST2_0_en[2][1:0]
d2a_DIST_ST2_1_en[2][1:0]
d2a_daisy_chain[4][2:0]
d2a_sw_2P1T[5:0]
d2a_center_bias_en[4:0]
d2a_quad_bias_en[3:0]
d2a_mulitB_fe_cbias[8][2:0]
d2a_PTAT_St1[8][5:0]
d2a_PTAT_St2[8][5:0]
d2a_PTAT_St3[8][5:0]
d2a_CTAT[8][5:0]
d2a_DIST_CTAT[5:0]
d2a_DIST_St1_PTAT[2][5:0]
d2a_DIST_St2_0_PTAT[2][5:0]
d2a_DIST_St2_1_PTAT[2][5:0]
d2a_DIST_St1_cbias[3][2:0]
d2a_DIST_St2_0_cbias[2][2:0]
d2a_DIST_St2_1_cbias[2][2:0]
d2a_captune[8][7:0]

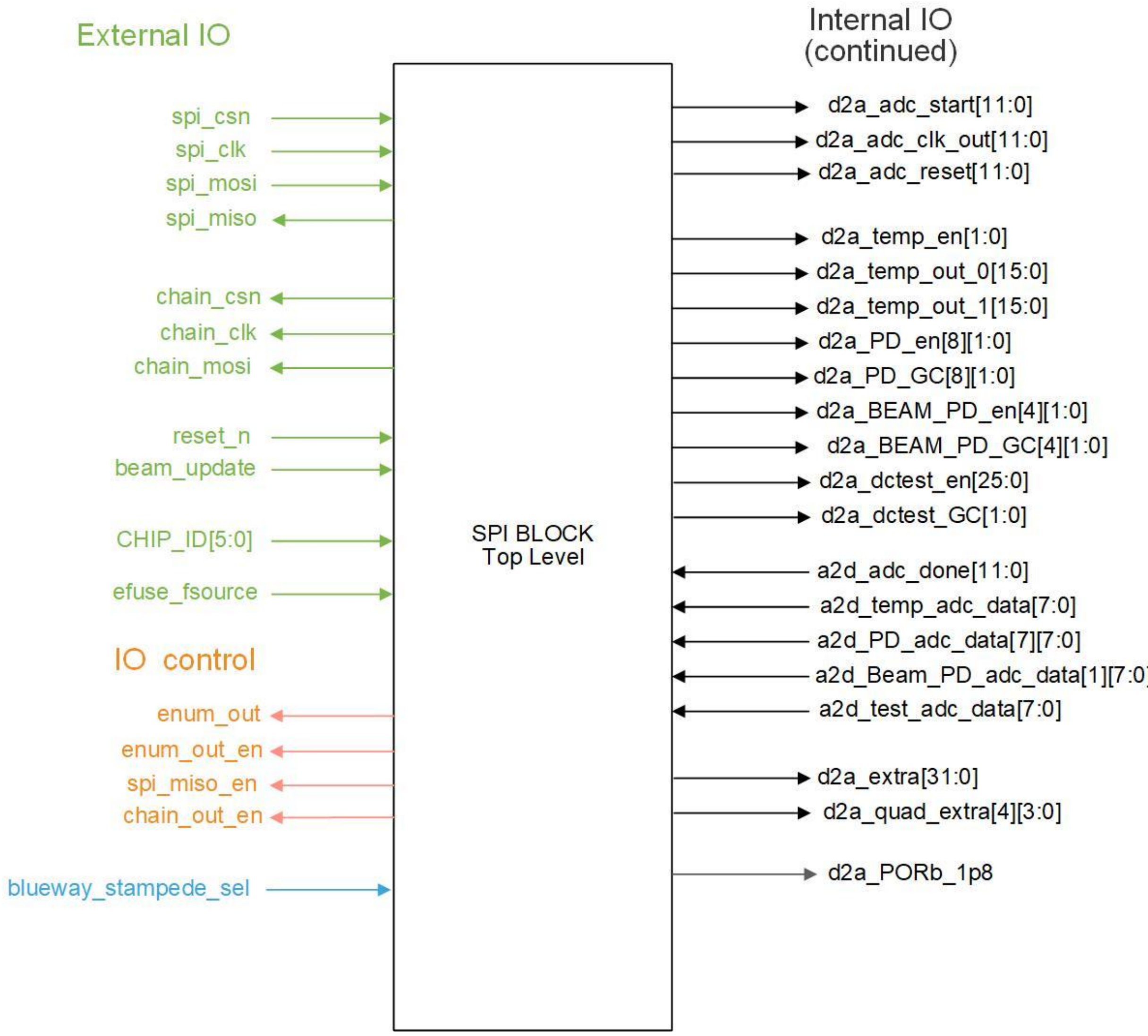


Figure 12: Internal Interface

Pin Name	Direction	Connection	Function
SPI_CSN	In	external pad	SPI chip select, active low
SPI_CLK	In	external pad	SPI clock
SPI_MOSI	In	external pad	SPI data in
SPI_MISO	out	external tri-state	SPI data out
RESET_N	In	external pad	block reset, active low
BEAM_UPDATE	In	external pad	update beam set latch
CHAIN_CSN	Out	external pad	Daisy chain output of SPI_CSN
CHAIN_CLK	out	external pad	Daisy chain output of SPI_CLK
CHAIN_DATA	out	external pad	Daisy chain output of SPI_MOSI
CHIP_ID[5:0]	In	external pad	6-bit chip id

blueway_stampede_sel	In	Internal	Indicator from chip top level for Blueway VS Stampede
D2a_phase_out[8][2][8:0]	out	internal	Phase control output 9-bits per beam/channel
D2a_atten_out[8][2][6:0]	out	internal	Atten control output 7-bits per beam/channel
D2a_atten_cal[8][2][3:0]	out	internal	Atten calibration 4-bits per beam/channel
D2a_FE_atten[8][3:0]	Out	Internal	Front End Attenuator 4-bits per channel
D2a_beam_atten[2][5:0]	Out	Internal	Common attenuator 6-bits per beam
Gain_control[8][2:0]	out	internal	Per channel gain control 3-bits per channel
Fe_Bias_bits[8][23:0]	out	internal	Amplifier bias control bits 24-bits per channel
Common_bias_bits[2][11:0]	Out	Internal	Common amplifier bias control 12-bits per beam
Enables/PD_gain[4][15:0]	Out	Internal	Enables for power detectors, temp sensors, band gap references, and gain control setting for PD 16-bits per quad
captune_bits[4[11:0]	Out	Internal	Spare bits for tuning the Quads. 12-bits per quad
fe_pwr_out[7:0]	out	internal	LNA power on state per channel
fe_sw_out[8][5:0]	out	internal	combiner switch state 2-bits per beam
common_pwr_out[2:0]	out	internal	Common power on state for beams
common_sw_out[5:0]	out	internal	Common switch state, 2 bits per beam
adc_start[11:0]	out	internal	conversion start pulse for adcs
adc_done[11:0]	In	internal	adc conversion done handshake
adc_input_vec[95:0]	In	internal	adc input data, 8 bits x 13 ADCs
Temp_out_offset[14:0]	Out	Internal	Temperature sensor offset calibration
Temp_out_slope[14:0]	Out	Internal	Temperature sensor slope calibration

Table 8: Block Pin Descriptions

Below is the functional block diagram for the SPI interface.

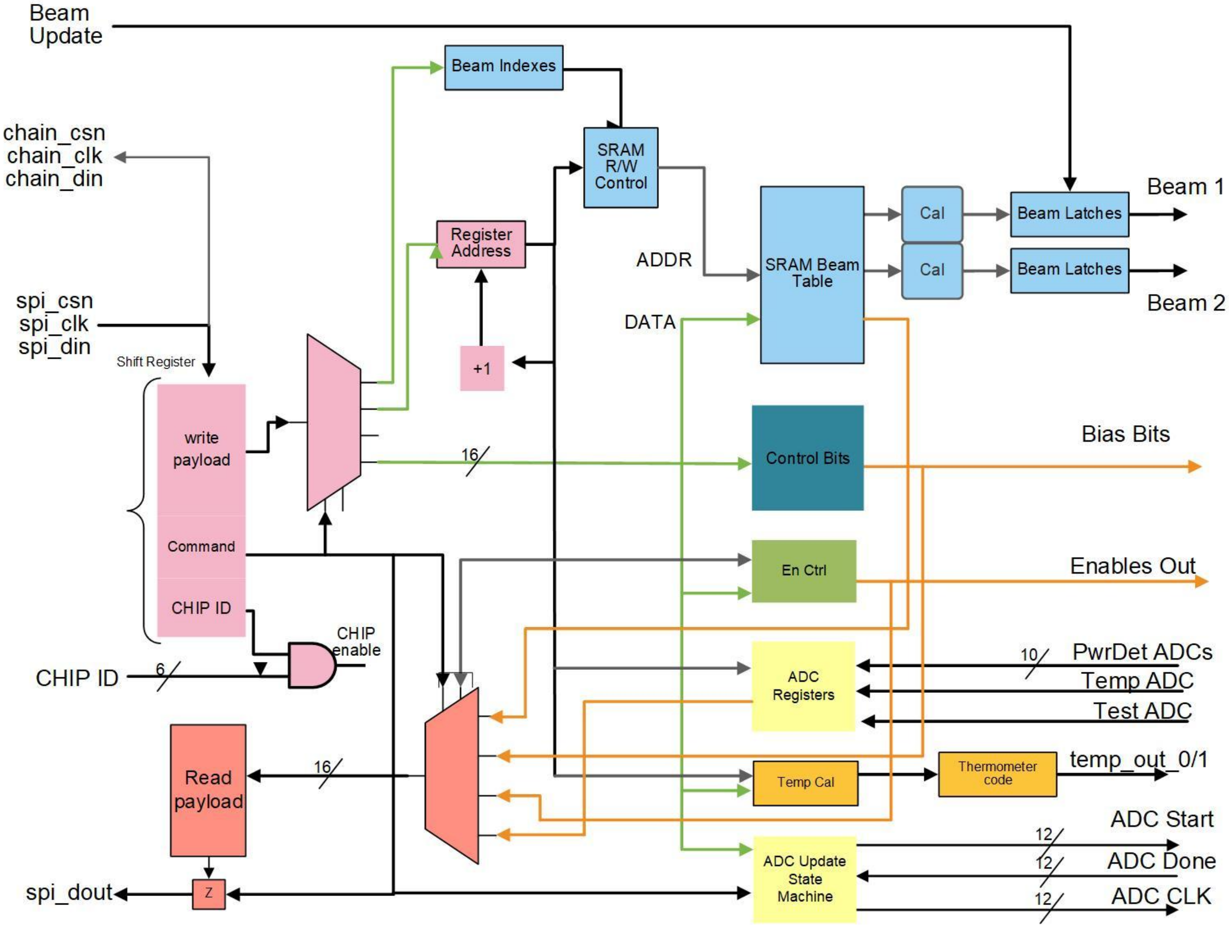


Figure 13: SPI Block Diagram

SPI write/load state machine

The SPI interface has an internal state machine for handling the data writes. This state machine will count SPI_CLK and generate a pulse on the final clock of the payload. This pulse will remain high until the next clock or the SPI_CSN goes high. The falling edge of SPI_CLK is used to capture the data into the register file when the end of transaction pulse is valid. It is expected that SPI_CSN goes high after the falling edge of SPI_CLK to complete the transaction. In general, it is assumed that the SPI_CLK is not continuous and will stop at the end of the payload, but this is not required if the correct timing relationship is maintained with respect to the SPI_CSN.

SPI Read Timing

The counter in the SPI write state machine is also used to control the read function. The output shift register is loaded with the register pointed to by the address pointer during the CHIP_ID and command portion of the SPI transaction. Once that phase is completed, assuming the CHIP_ID matches and the command is for a read operation, the SPI_OUT_EN is set high to enable the first bit of the output shift register to be passed through the tri-state buffer. On each subsequent clock the shift register is updated, and the next bit is output. The figure below shows the timing.

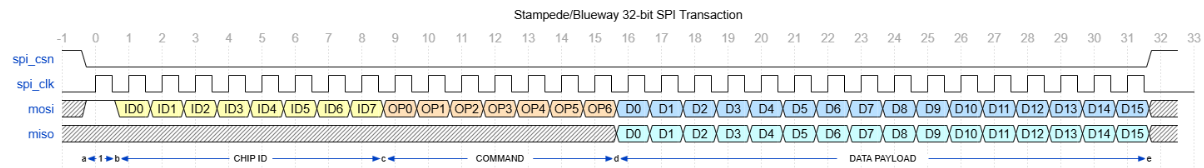


Figure 14: SPI Read Timing

Beam Updating

The BEAM_UPDATE signal provides a mechanism for a synchronized update of the beam forming parameters of all quads and both polarizations. There are four beam set pointers two beams per quad channel. Each pointer can be updated individually to set a line in the beam table storage. The update is done through a SPI transaction to write a set pointer register, then when the BEAM_UPDATE signal is pulsed, the sets are used to update the beamforming parameters output to the analog circuits.

Note the latch updates for the beam_set are asynchronous to the SPI_CLK and SPI_CSN, but to ensure antenna pattern integrity the latches should only be pulsed when the SPI_CSN is high. Additionally, writing to a beam_table address which is selected by an active beam set pointer will immediately change the phase and attenuation for that quadrant at the end of the transaction. This is not recommended for normal operation as it takes multiple operations to update the parameters of all elements in an array and it can lead to an undesired transient beam pattern.

Is the pulse_en bit is set then the enables for the beam amplifiers are updated based on the BEAM_UPDATE pin as shown in the figure below.

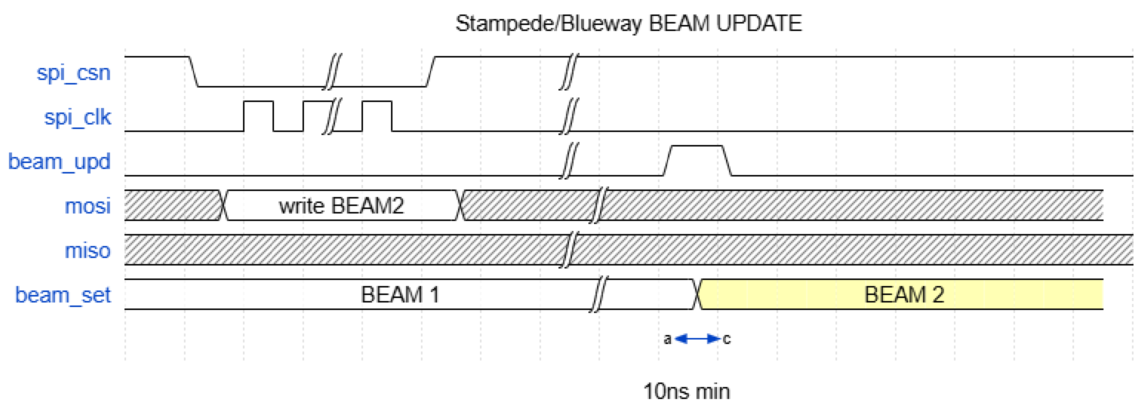


Figure 15: Beam Update Timing

ADC Interface

The ADC interface from the SPI block is designed to be compatible with an 8-bit SAR type ADC. The block will provide a start signal to the ADC to initiate conversion. The SPI_CLK is used for the SAR process and then a done signal is pulsed by the ADC when the conversion is done. The done signal must be coincident with the output data and occurs 24 ADC clocks after the start signal. The ADC should not expect the clock to be active before the start or after a done. The handshaking is shown below.

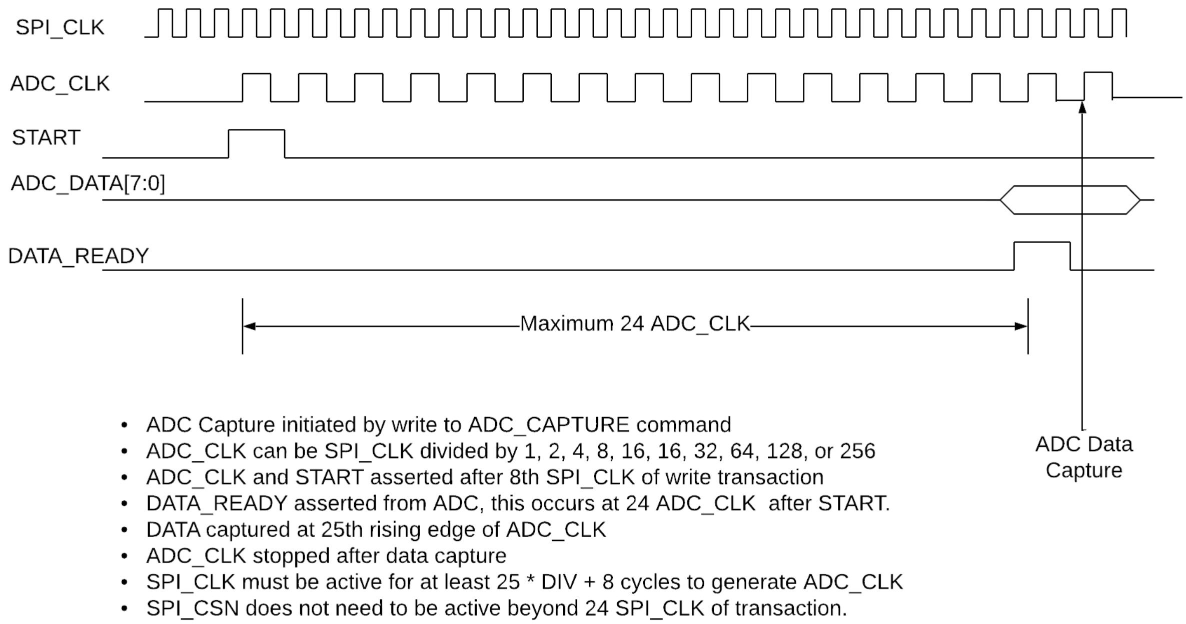


Figure 16: ADC timing

The ADC clock can be divided down from the SPI_CLK by 2^N for N = 0-8. The SPI_CLK must be active during the ADC_CAPTURE transaction but SPI_CSN should not extend beyond the 32 cycles of the transaction.

Reset Operation

The RESET_N signal will reset the device to a default state and cause the calibration data to be read from the efuse. To accomplish this the SPI_CLK must be active for approximately 1200 cycles after the reset is released. During this time the efuse will be read into the register space and then a soft reset applied to load the calibration bits from the efuse registers. Each Quad reads local efuse to populate the Phase_calibration_Luts. Efuse bits for atten_cal, phase_offset, tuning_bits, biases, and PD settings are read from the center block efuse and distributed to the Quad registers via and internal serial link.

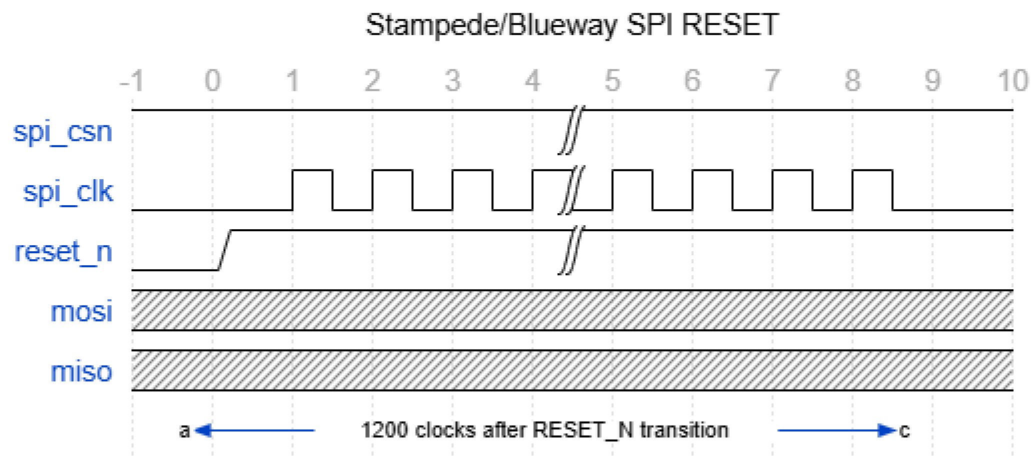


Figure 17: Reset Timing

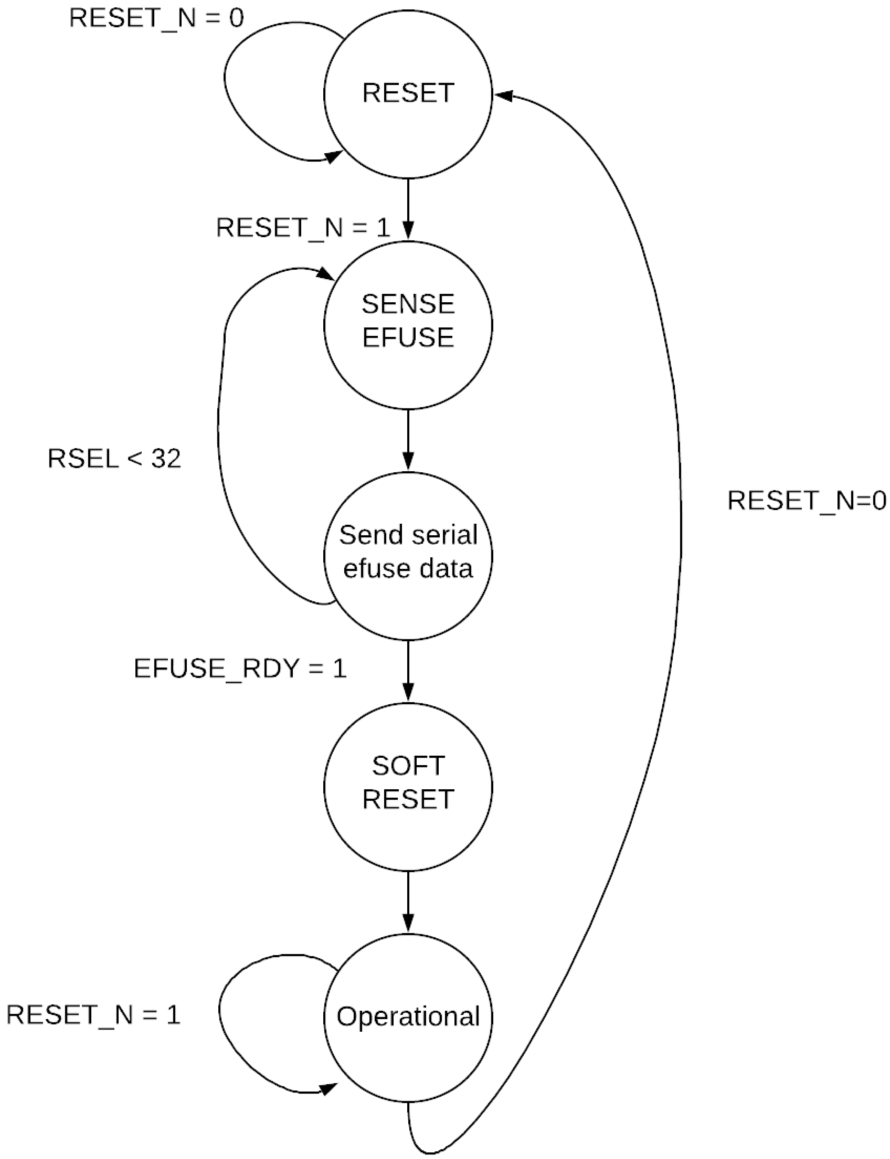


Figure 18: Reset State Machine

Auto Enumeration

The SPI interface supports automatic enumeration of the CHIP_ID values. This allows for assigning CHIP_ID values independently of the CHIP_ID pins on the device and setting CHIP_ID values greater than 64. The enumeration process assumes the CHIP_ID pins of the devices in the array are connected in a daisy chain where CHIP_ID[1] (enumeration output) is connected to CHIP_ID[0] (enumeration input) of the subsequent chip, as shown in Figure 9. The CHIP_ID[0] of the first device in the array can be tied high or driven by the SPI controller to a high state during enumeration.

Enumeration is initialized by sending a SPI transaction with the INIT_ENUM command. This should be sent to all devices in the array simultaneously using broadcast addressing (CHIP_ID = 0xff). On the subsequent SPI transaction the device will count the number of SPI_CLKs until it detects CHIP_ID[0] pin is high. It will transfer the clock count to the SHORT_ID of the device and output the enumeration pulse on CHIP_ID[1], thus the devices will have SHORT_IDs assigned in order of the daisy chain sequence. The timing diagram for the enumeration process is shown in Figure 19.

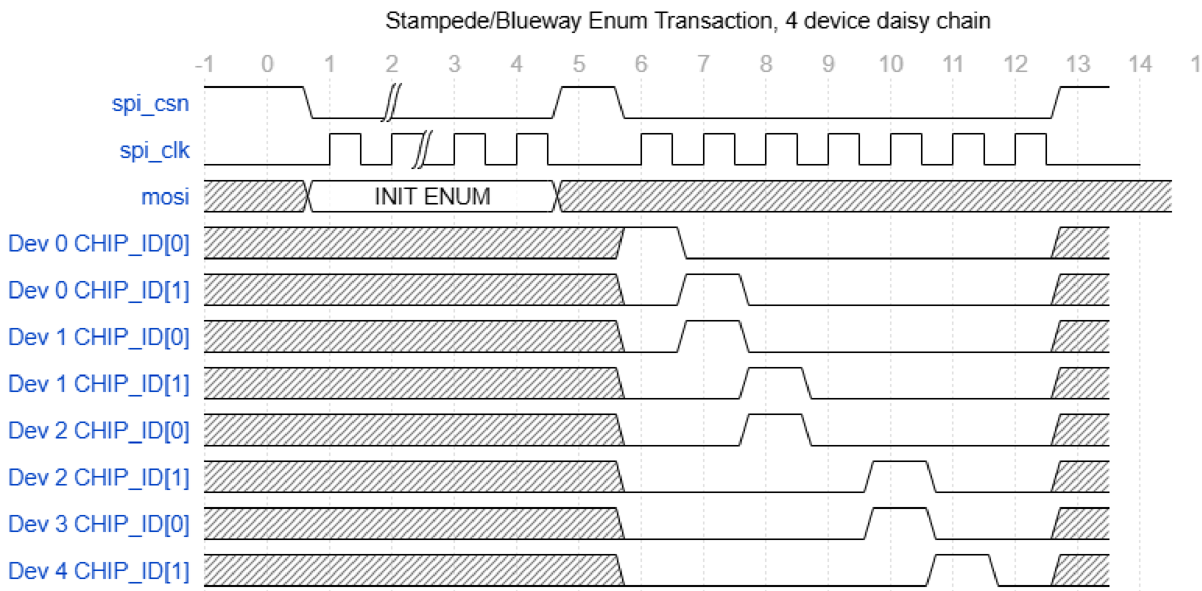


Figure 19: Enumeration Timing

Error Handling

The SPI interface will recover from error conditions related to few or too many clocks per chip-select phase. The internal state machine resets to a waiting state whenever SPI_CSN is high. If the rising edge of SPI_CSN occurs before 32 clocks, then no write or load action will be performed, and the command is discarded (note some command actions which do not have data associated with them are initiated after 16 clocks and will be executed on a short transaction). If a read command is being executed, then it will terminate when SPI_CSN goes high. If this occurred with less than 32 clocks then the remaining bits of the payload will not be output, if it occurs after 32 clocks then it will be treated as an extended read and the next word in the register space will be shifted out. A write or load command will be executed on the falling edge of SPI_CLK after 32 clocks. If SPI_CSN remains low after 32 clocks the state machine counter will wrap around, and the interface goes into extended transaction mode where each subsequent set of 16-bits is treated as a new payload for the current command.

Efuse Programming

There is a set of commands available for programming the efuse of the device. The efuse is arranged into 16 rows of 32 bits. Programming is accomplished 1 row at a time. The process for programming is shown in the flow chart below. It involves loading the programming data register and the row selection register then issuing the EFZ_PROG_CMD. Programming takes up to 10 usec per non-zero bit in the data register. There is a status bit which can be polled to detect when the programming operation is complete. Once complete the data should be read to verify programming then the next row can be programmed.

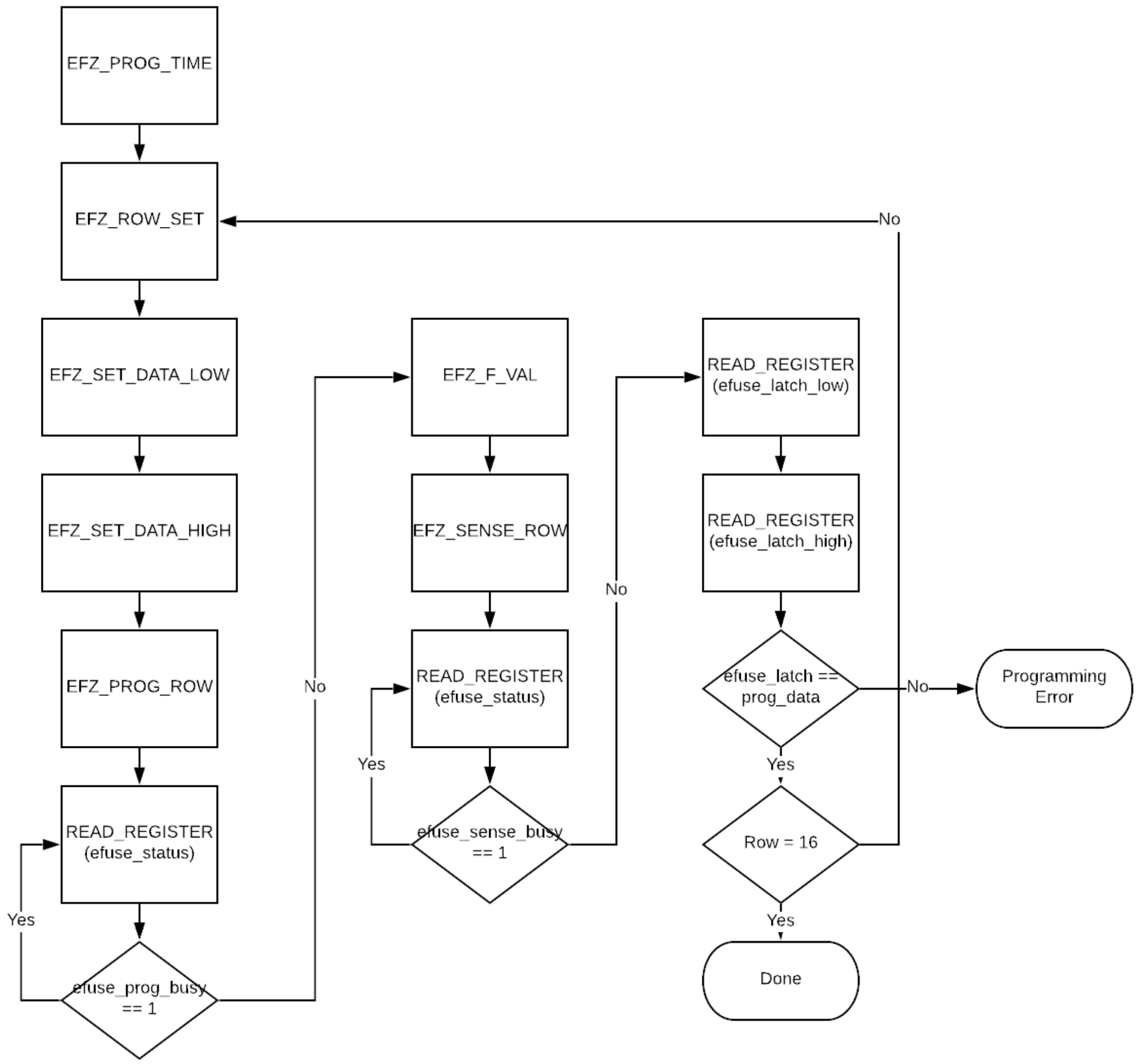


Figure 20: Efuse Programming Flow

The 45RFSOI_EFUSE512_databook from Global Foundries should be referenced for details about the Efuse programming time and sense F-value.

Internal Architecture

Although the SPI interface is presented as a single design it is implemented as two sub-blocks. There is a central controller which instantiates the efuse IP and Quad block which contains the beam table storage for 1 quadrant of the chip. Figure 21 shows the internal structure of the SPI interface.

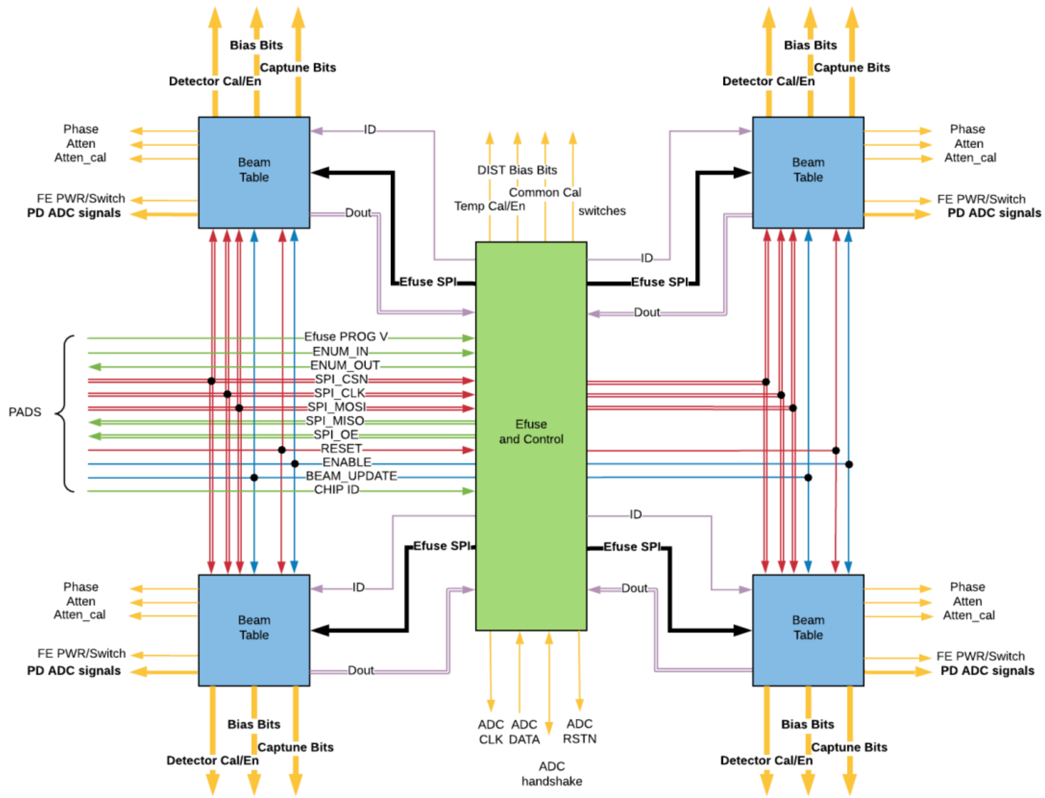


Figure 21: Internal Architecture

Register Definitions

BEAM	Read/write	These registers contain the beam former parameters for a beam index address. Each line in the beam table contains 4 16-bit words, 1 for each quadrant. Each 16 bits word has a 7 bit attenuator value (32 dB range with 0.25 dB step size), and a 7-bit phase setting (360 degree range with 2.8 degree step size). The values are not initialized on reset.
CHIP ID	Read only	This register holds the current state of the CHIP_ID used to filter SPI transactions. Reset Value from external pins
VERSION	Read Only	This register holds the silicon version information for the chip. The current value of this register is 0 for Blueway and 1 for Stampede.
UID	Read only	These two registers contain the 32-bit unique ID programmed into the chip during factory calibration. Reset value from calibration
DAISY CHAIN AMP BYPASS	Read/write	This register contains the control for the 4 daisy chain amplifiers. The control bits select if the amp is bypassed. Reset Value = 1111

DAISY CHAIN AMP THRU	Read/write	This register contains the control for the 4 daisy chain amplifiers. The control bits select if the amp passes the signal through. Reset Value = 0000.
DAISY CHAIN AMP PWRDN	Read/write	This register contains the control for the 4 daisy chain amplifiers. The control bits select if the amp is in the power down state. 0 = power down, 1 = power up. Reset Value = 0000.
DAISY CHAIN AMP BIAS PWRDN	Read/write	This register contains the control for the 4 daisy chain amplifiers. The control bits select if the amp bias is powered down. 0 = power down, 1 = power up. Reset Value = 0000.
COMMON GAIN	Read/write	This register sets the common mode attenuation for each of the beams after combining. The range is 0-63 in 0.5 dB steps. Reset Value = 0x0000
PULSE_EN_1	Read/Write	This bit selects if the COMMON GAIN values are latch enabled with the BEAM_UPDATE signal. 1 = Enable latch, 0 = immediate effect. Reset value = 0.
CENTER ENBG	Read/Write	This bit enables the center block band gap reference. 1 = enable, 0 = disable. Reset value = 0.
CENTER ENREG	Read/Write	This bit enables the center block LDO. 1 = enable, 0 = disable. Reset value = 0.
CENTER PD OVERRIDE	Read/Write	This bit overrides the auto power down control for the common amplifiers, 1 = power down override, 0 = auto power down. Reset value = 1.
SPI DAISY CHAIN DISABLE	Read/Write	This bit will disable the SPI DAISY CHAIN outputs, 1 = Disable, 0 = Enable. Reset value = 0.
GLOBAL POWER DOWN	Read/Write	This bit will put all the amplifiers into power down state if in auto power down mode, 1 = power down, 0 = power up. Reset value = 1.
PULSE_EN_2	Read/Write	This bit controls if the following bits are latch enabled with the BEAM_UPDATE signal: BEAM PWRDN, BEAM ENABLES, BEAM MATCH, GLOBAL POWER DOWN, SPI DAISY CHAIN DISABLE, CENTER PD OVERRIDE, CENTER BIAS ENABLE. 1 = Enable latch, 0 = immediate effect. Reset value = 0.
DS_DOUT	Read/Write	These bits set the drive strength for SPI_DOUT: 00 = 2 mA, 01 = 4 mA, 10 = 8 mA, 11 = 12 mA. Reset value = 11
DS_CHAIN	Read/Write	These bits set the drive strength for SPI_DAISY_CHAIN outputs:

		00 = 2 mA, 01 = 4 mA, 10 = 8 mA, 11 = 12 mA. Reset value = 11
CENTER EFUSE PWRDN DISABLE	Read/Write	This bit will override the auto power down feature of the efuse. 1 = power down disable, 0 = power down enabled. Reset value = 0.
BEAM DIST Enables	Read/Write	These bits enable the distribution amplifiers for each beam. 1 = enabled, 0 = disabled. Reset value = 0.
CENTER BIAS BEAM ENABLE	Read/Write	This bit enables the bias generation per beam. 1 = bias generation enabled, 0 = bias generation disabled. Reset value = 0.
BEAM PWRDN	Read/Write	These bits control the power down state of the beam common amplifiers when in PD override mode. 0 = power down, 1 = power up. Reset value = 0.
BEAM MATCH	Read/Write	This bit controls the matching state of the beam output in Blueway. 1 = disable, 0 = enable. Reset = 0.
CHx ENABLES	Read/Write	These bits control the per channel beam enables. There are individual bits for each beam former on each polarization. 1 = enable, 0 = disable. Reset value = 0.
PULSE EN CHx 1	Read/Write	This bit selects if the CHx ENABLE values are latch enabled with the BEAM_UPDATE signal. 1 = Enable latch, 0 = immediate effect. Reset value = 0.
CHx PWRDN	Read/Write	These bits control the power down state of the Quad amplifiers for Quad x. 0 = power down, 1 = power on. Reset value = 0.
CHx PD OVERRIDE	Read/Write	This bit overrides the auto power down control for the Quad x amplifiers, 1 = power down override, 0 = auto power down. Reset value = 1.
CHx BIAS EN	Read/Write	This bit enables the bias generation for Quad x. 1 = bias generation enabled, 0 = bias generation disabled. Reset value = 0.
PULSE EN CHx 1	Read/Write	This bit selects if the following values are latch enabled with the BEAM_UPDATE signal: CHx PWRDN CHx PD OVERRIDE CHx BIAS EN 1 = Enable latch, 0 = immediate effect. Reset value = 0.
CHx EFUSE PWRDN DISABLE	Read/Write	This bit will override the auto powerdown feature of the efuse for Quad x. 1 = power down disable, 0 = power down enabled. Reset value = 0.
CHx CAL FREQ SEL BEAMy	Read/write	These bits selects the frequency range for which the calibration is optimized. Reset Value = 0
PULSE EN CHx 2	Read/Write	This bit selects if the CHx CAL FREQ SEL values are latch enabled with the BEAM_UPDATE signal.

		1 = Enable latch, 0 = immediate effect. Reset value = 0.
GAIN CONTROL x	Read/write	These bits set the front end attenuator in Blueway for the H and V channels of Quad x. The control is 8dB attenuation for each bit. Bit 0 for LNA, Bit 1 for ST2 LNA, Bit 2 for ST3 LNA, and Bit 3 is reserved. Reset Value = 0
PULSE EN CHx 3	Read/Write	This bit selects if the GAIN CONTROL x values are latch enabled with the BEAM_UPDATE signal. 1 = Enable latch, 0 = immediate effect. Reset value = 0.
TEMP CAL/CONTROL	Read/write	This register set the temp sensor calibration and enable. The calibration should be factory preset so only the enable bits need to be set to 1 if the temp sensor is to be used. Reset Value from calibration
PD CONTROL QUAD x	Read/write	These registers have the enables bits for the power detectors. They should be set to 1 if the power detectors are to be used. Reset Value = 0
DCTEST EN QUADx	Read/write	These bits select the DC value source into the DCTEST ADC for Quad x. See table 12 below. Reset value = 0
PD GAIN QUADx	Read/write	This register controls the gain for each of the power detectors. Reset Value from calibration
ADC CLK DIV	Read/write	This register sets the clock divider for the ADC conversion clock. This register should be set to 1 for normal operation. Reset Value = 1
ADC CLK POL	Read/write	This register inverts the polarity of the ADC clock with respect to the SPI_CLK. This is a debug feature. Reset value = 0.
ADC MASK	Read/write	This register sets the enable mask for the ADCs. If a bit is set to 0 then the corresponding ADC will not execute a conversion in response to an ADC_CAPTURE command. Reset Value = 0x3FFF
ADC RESET POL	Read/write	This register inverts the polarity of reset signal applied to the ADCs with respect to the reset pin. Toggling this pin can provide a soft reset of the ADCs. This bit should be 1 for normal operation. Reset value = 0.
ADC AUTO RESET DISABLE	Read/write	This bit will disable the auto reset applied to the ADCs at the start of each capture operation. This is a debug feature. 1 = disable, 0 = enable. Reset value = 0.
ADC	Read only	These registers contain the 8-bit values for the most recent ADC_CAPTURE operation. Reset Value = 0x0000
DCTEST EN CENTR	Read/Write	These bits select the DC source multiplexed into the DCTEST ADC from the center block. See Table 12 below. Reset value = 0

DCTEST ENBGR	Read/Write	This enables the bandgap reference for the DCTEST ADC. 1 = enable, 0 = disable. Reset value = 0.
DCTEST GC	Read/Write	This enables the gain control for the DCTEST ADC. Reset value = 0
DCTEST MODE	Read/Write	These bits select the Mode from the DCTEST source for the DCTEST ADC. See table 13 below. Reset value = 0.
BIAS	Read/Write	These registers program the bias current for various amplifiers. See table 10 below. Reset value from calibration.
DAISY CHAIN BIAS	Read/Write	These register control the Daisy Chain Amplifier bias and tunning. See table 10 below. Reset value from calibration.
EXTRA	Read/Write	These bits are extra register bits. See table 11 below for function. Reset value 0
xBy ATTEN CAL	Read/Write	These bits set the beam calibration attenuator for beam y of Quad x. They have 4dB attenuation range with 0.25 dB step size. Reset value from calibration.
Qx CAPTUNE	Read/Write	These bits are tuning bits for each quad. See table 11 below for function. Reset value from calibration.
PHASE CALIBRATION	Read/Write	These bits hold the calibration data for the RTPS phase control. Reset value from calibration.
ADC DEBUG	Read only	These registers return the per Quad settings for the ADC CLK and MASK registers. For debug only.
BEAM y INDEX QUADx	Read only	These registers return the current beam pointers for each beam in each quad. For debug only.
BEAM y SET QUADx	Read only	These registers return the current attenuator and calibrated phase settings output for each beam and each quad. For debug only.
EFUSE QUAD DEBUG	Read only	These registers return the EFUSE control register contents, below, from each QUAD for debug purposes.
EFUSE DATA	Read only	These registers return the EFUSE content read by the center block as raw bits. For debug only.
EFUSE ROW	Read only	The current row pointer for EFUSE reading or writing. For debug only.
EFUSE PROG CNT	Read only	The timer value used for EFUSE programming to insure 10usec programming interval. For debug only.
EFUSE F VAL	Read only	Resistance value for EFUSE sense. For debug only.
EFUSE STATUS	Read only	EFUSE state machine status and test parameters. For debug only.
EFUSE PROG DATA	Read only	32 bit data to be programmed into EFUSE. For debug only.
EFUSE SENSE	Read only	The last 32 bit value read from the EFUSE in the center block. For debug only.

Table 9: Register descriptions

Bias Mappings and Tuning Bits

Table 10 below indicates the register address and bits with the register for amplifier bias controls.

Front End Amplifiers per QUAD						
Channel	**PTAT_ST 1**	**PTAT_ST 2**	**PTAT_ST 3**	**CTAT**	**MultB_fe_cbias**	
CH0 H-pol	0x1038 bits 0-5	0x1038 Bits 8-13	0x103C Bits 0-5	0x103C bits 8-13	0x1048 Bits 0-2	
CH0 V-pol	0x1040 bits 0-5	0x1040 Bits 8-13	0x1044 Bits 0-5	0x1044 bits 8-13	0x1048 Bits 8-10	
CH1 H-pol	0x1039 Bits 0-5	0x1039 Bits 8-13	0x103D Bits 0-5	0x103D Bits 8-13	0x1049 Bits 0-2	
CH1 V-pol	0x1041 Bits 8-13	0x1041 Bits 8-13	0x1045 Bits 0-5	0x1045 Bits 8-13	0x1049 Bits 8-10	
CH2 H-pol	0x103A Bits 0-5	0x103A Bits 8-13	0x103E Bits 0-5	0x103E bits 8-13	0x104A Bits 0-2	
CH2 V-pol	0x1042 Bits 0-5	0x1042 Bits 8-13	0x1046 Bits 0-5	0x1046 bits 8-13	0x104A Bits 8-10	
CH3 H-pol	0x103B Bits 8-13	0x103B Bits 8-13	0x103F Bits 0-5	0x103F Bits 8-13	0x104B Bits 0-2	
CH3 V-pol	0x1043 Bits 8-13	0x1043 Bits 8-13	0x1047 Bits 0-5	0x1047 Bits 8-13	0x104B Bits 8-10	
Post Combining Amplifiers per BEAM						
Beam	**PTAT_ST 1**	**PTAT_ST2_0**	**PTAT_ST2_1**	**Cbias ST1**	**Cbias ST2_0**	**Cbias ST2_1**
DIST B0	0x104C Bits 0-5	0x104C Bits 8-13	0x104D Bits 0-5	0x104D Bits 6-8	0x104D Bits 9-11	0x104D Bits 12-14
DIST B1	0x104E Bits 0-5	0x104E Bits 8-13	0x104F Bits 0-5	0x104F Bits 6-8	0x104F Bits 9-11	0x104F Bits 12-14
DIST CTAT	0x1052 Bits 0 -5					

Table 10: Bias Register Locations

Quad/ Center/ Mixed	**Bit name**	**bits**	**RX**	**TX**	**Blueway Function**	**Stampede Function**
Quad0	d2a_CH0_H_PTAT_St1	6	yes	yes	PTAT current bias of St1 LNA for CH0 H	PTAT current bias of PA for CH0 H
Quad0	d2a_CH0_H_PTAT_St2	6	yes	yes	PTAT current bias of St2 LNA for CH0 H	PTAT current bias of PA Driver for CH0 H
Quad0	d2a_CH0_H_PTAT_St3	6	yes	yes	PTAT current bias of St3 splitter for CH0 H	PTAT current bias of combiner for CH0 H
Quad0	d2a_CH0_V_PTAT_St1	6	yes	yes	PTAT current bias of St1 LNA for CH0 V	PTAT current bias of PA for CH0 V
Quad0	d2a_CH0_V_PTAT_St2	6	yes	yes	PTAT current bias of St2 LNA for CH0 V	PTAT current bias of PA Driver for CH0 V
Quad0	d2a_CH0_V_PTAT_St3	6	yes	yes	PTAT current bias of St3 splitter for CH0 V	PTAT current bias of combiner for CH0 V
Quad1	d2a_CH1_H_PTAT_St1	6	yes	yes	PTAT current bias of St1 LNA for CH1 H	PTAT current bias of PA for CH1 H

Quad1	d2a_CH1_H_PTAT_St2	6	yes	yes	PTAT current bias of St2 LNA for CH1 H	PTAT current bias of PA Driver for CH1 H
Quad1	d2a_CH1_H_PTAT_St3	6	yes	yes	PTAT current bias of St3 splitter for CH1 H	PTAT current bias of combiner for CH1 H
Quad1	d2a_CH1_V_PTAT_St1	6	yes	yes	PTAT current bias of St1 LNA for CH1 V	PTAT current bias of PA for CH1 V
Quad1	d2a_CH1_V_PTAT_St2	6	yes	yes	PTAT current bias of St2 LNA for CH1 V	PTAT current bias of PA Driver for CH1 V
Quad1	d2a_CH1_V_PTAT_St3	6	yes	yes	PTAT current bias of St3 splitter for CH1 V	PTAT current bias of combiner for CH1 V
Quad2	d2a_CH2_H_PTAT_St1	6	yes	yes	PTAT current bias of St1 LNA for CH2 H	PTAT current bias of PA for CH2 H
Quad2	d2a_CH2_H_PTAT_St2	6	yes	yes	PTAT current bias of St2 LNA for CH2 H	PTAT current bias of PA Driver for CH2 H
Quad2	d2a_CH2_H_PTAT_St3	6	yes	yes	PTAT current bias of St3 splitter for CH2 H	PTAT current bias of combiner for CH2 H
Quad2	d2a_CH2_V_PTAT_St1	6	yes	yes	PTAT current bias of St1 LNA for CH2 V	PTAT current bias of PA for CH2 V
Quad2	d2a_CH2_V_PTAT_St2	6	yes	yes	PTAT current bias of St2 LNA for CH2 V	PTAT current bias of PA Driver for CH2 V
Quad2	d2a_CH2_V_PTAT_St3	6	yes	yes	PTAT current bias of St3 splitter for CH2 V	PTAT current bias of combiner for CH2 V
Quad3	d2a_CH3_H_PTAT_St1	6	yes	yes	PTAT current bias of St1 LNA for CH3 H	PTAT current bias of PA for CH3 H
Quad3	d2a_CH3_H_PTAT_St2	6	yes	yes	PTAT current bias of St2 LNA for CH3 H	PTAT current bias of PA Driver for CH3 H
Quad3	d2a_CH3_H_PTAT_St3	6	yes	yes	PTAT current bias of St3 splitter for CH3 H	PTAT current bias of combiner for CH3 H
Quad3	d2a_CH3_V_PTAT_St1	6	yes	yes	PTAT current bias of St1 LNA for CH3 V	PTAT current bias of PA for CH3 V
Quad3	d2a_CH3_V_PTAT_St2	6	yes	yes	PTAT current bias of St2 LNA for CH3 V	PTAT current bias of PA Driver for CH3 V
Quad3	d2a_CH3_V_PTAT_St3	6	yes	yes	PTAT current bias of St3 splitter for CH3 V	PTAT current bias of combiner for CH3 V
Quad0	d2a_CH0_H_CTAT	6	no	yes	reserved bits	<3:1> to efuse; CTAT current with 15.626uA step size and the range from 0 to 1000uA.
Quad0	d2a_CH0_V_CTAT	6	no	yes	reserved bits	<3:1> to efuse; CTAT current with 15.626uA step size and the range from 0 to 1000uA.
Quad1	d2a_CH1_H_CTAT	6	no	yes	reserved bits	<3:1> to efuse; CTAT current with 15.626uA step size and the range from 0 to 1000uA.
Quad1	d2a_CH1_V_CTAT	6	no	yes	reserved bits	<3:1> to efuse; CTAT current with 15.626uA step size and the range from 0 to 1000uA.
Quad2	d2a_CH2_H_CTAT	6	no	yes	reserved bits	<3:1> to efuse; CTAT current with 15.626uA step size and the range from 0 to 1000uA.
Quad2	d2a_CH2_V_CTAT	6	no	yes	reserved bits	<3:1> to efuse; CTAT current with 15.626uA step size and the range from 0 to 1000uA.
Quad3	d2a_CH3_H_CTAT	6	no	yes	reserved bits	<3:1> to efuse; CTAT current with 15.626uA step size and the range from 0 to 1000uA.
Quad3	d2a_CH3_V_CTAT	6	no	yes	reserved bits	<3:1> to efuse; CTAT current with 15.626uA step size and the range from 0 to 1000uA.
Quad0	d2a_CH0_H_multB_fe_cbias	3	no	yes	reserved bits	<2:0> for the middle cascode device gate voltage of 3to1 Comb
Quad0	d2a_CH0_V_multB_fe_cbias	3	no	yes	reserved bits	<2:0> for the middle cascode device gate voltage of 3to1 Comb
Quad1	d2a_CH1_H_multB_fe_cbias	3	no	yes	reserved bits	<2:0> for the middle cascode device gate voltage of 3to1 Comb

Quad1	d2a_CH1_V_multB_fe_cbias	3	no	yes	reserved bits	<2:0> for the middle cascode device gate voltage of 3to1 Comb
Quad2	d2a_CH2_H_multB_fe_cbias	3	no	yes	reserved bits	<2:0> for the middle cascode device gate voltage of 3to1 Comb
Quad2	d2a_CH2_V_multB_fe_cbias	3	no	yes	reserved bits	<2:0> for the middle cascode device gate voltage of 3to1 Comb
Quad3	d2a_CH3_H_multB_fe_cbias	3	no	yes	reserved bits	<2:0> for the middle cascode device gate voltage of 3to1 Comb
Quad3	d2a_CH3_V_multB_fe_cbias	3	no	yes	reserved bits	<2:0> for the middle cascode device gate voltage of 3to1 Comb
Center	d2a_DIST_CTAT	6	no	yes	reserved bits	<3:1> to efuse; CTAT current with 15.626uA step size and the range from 0 to 1000uA.
Center	d2a_DIST_B0_St1_cbias	3	no	yes	reserved bits	<2:0> for DIST St1 middle casocde device gate bias;
Center	d2a_DIST_B1_St1_cbias	3	no	yes	reserved bits	<2:0> for DIST St1 middle casocde device gate bias;
Center	d2a_DIST_B0_St2_0_cbias	3	no	yes	reserved bits	<2:0> for DIST St2_0 and St2_1 middle casocde device gate bias;
Center	d2a_DIST_B1_St2_0_cbias	3	no	yes	reserved bits	<2:0> for DIST St2_0 and St2_1 middle casocde device gate bias;
Center	d2a_DIST_B0_St2_1_cbias	3	no	yes	reserved bits	<2:0> for DIST St2_0 and St2_1 bottom gm device gate bias;
Center	d2a_DIST_B1_St2_1_cbias	3	no	yes	reserved bits	<2:0> for DIST St2_0 and St2_1 bottom gm device gate bias;
Center	d2a_DIST_B0_St1_PTAT	6	yes	yes	PTAT current bias for the St2 combiner at Beam0	PTAT current bias for the St1 distribution network at Beam 0
Center	d2a_DIST_B1_St1_PTAT	6	yes	yes	PTAT current bias for the St2 combiner at Beam1	PTAT current bias for the St1 distribution network at Beam 1
Center	d2a_DIST_B2_St1_PTAT	6	yes	yes	<3:0> d2a_ptat_slope1<3:0>; <5:4> reserved bits;	<3:0> d2a_ptat_slope1<3:0>; <5:4> reserved bits;
Center	d2a_DIST_B0_St2_0_PTAT	6	yes	yes	reserved bits due to passive Wilk combiner	PTAT current bias for the CH1-0 St2 distribution network at Beam 0
Center	d2a_DIST_B1_St2_0_PTAT	6	yes	yes	reserved bits due to passive Wilk combiner	PTAT current bias for the CH1-0 St2 distribution network at Beam 1
Center	d2a_DIST_B2_St2_0_PTAT	6	yes	yes	<3:0> d2a_ptat_slope2<3:0>; <5:4> reserved bits;	<3:0> d2a_ptat_slope2<3:0>; <5:4> reserved bits;
Center	d2a_DIST_B0_St2_1_PTAT	6	yes	yes	reserved bits due to passive Wilk combiner	PTAT current bias for the CH3-2 St2 distribution network at Beam 0
Center	d2a_DIST_B1_St2_1_PTAT	6	yes	yes	reserved bits due to passive Wilk combiner	PTAT current bias for the CH3-2 St2 distribution network at Beam 1
Center	d2a_DIST_B2_St2_1_PTAT	6	yes	yes	<3:0> d2a_ptat_slope3<3:0>; <5:4> reserved bits;	<3:0> d2a_ptat_slope3<3:0>; <5:4> reserved bits;
Quad0	d2a_CH0_H_captune	8	yes	yes	Capacitance tune calibration for CH0 H; <3:0> for the St2 LNA and <7:4> for the St3 splitter;	Capacitance tune calibration for CH0 H; <2:0> for the combiner input and <4:3> for the combiner output; <7:5> reserved bits;
Quad0	d2a_CH0_V_captune	8	yes	yes	Capacitance tune calibration for CH0 V; <3:0> for the St2 LNA and <7:4> for the St3 splitter;	Capacitance tune calibration for CH0 V; <2:0> for the combiner input and <4:3> for the combiner output; <7:5> reserved bits;
Quad1	d2a_CH1_H_captune	8	yes	yes	Capacitance tune calibration for CH1 H; <3:0> for the St2 LNA and <7:4> for the St3 splitter;	Capacitance tune calibration for CH1 H; <2:0> for the combiner input and <4:3> for the combiner output; <7:5> reserved bits;
Quad1	d2a_CH1_V_captune	8	yes	yes	Capacitance tune calibration for CH1 V; <3:0> for the St2 LNA and <7:4> for the St3 splitter;	Capacitance tune calibration for CH1 V; <2:0> for the combiner input and <4:3> for the combiner output; <7:5> reserved bits;

Quad2	d2a_CH2_H_captune	8	yes	yes	Capacitance tune calibration for CH2 H; <3:0> for the St2 LNA and <7:4> for the St3 splitter;	Capacitance tune calibration for CH2 H; <2:0> for the combiner input and <4:3> for the combiner output; <7:5> reserved bits;
Quad2	d2a_CH2_V_captune	8	yes	yes	Capacitance tune calibration for CH2 V; <3:0> for the St2 LNA and <7:4> for the St3 splitter;	Capacitance tune calibration for CH2 V; <2:0> for the combiner input and <4:3> for the combiner output; <7:5> reserved bits;
Quad3	d2a_CH3_H_captune	8	yes	yes	Capacitance tune calibration for CH3 H; <3:0> for the St2 LNA and <7:4> for the St3 splitter;	Capacitance tune calibration for CH3 H; <2:0> for the combiner input and <4:3> for the combiner output; <7:5> reserved bits;
Quad3	d2a_CH3_V_captune	8	yes	yes	Capacitance tune calibration for CH3 V; <3:0> for the St2 LNA and <7:4> for the St3 splitter;	Capacitance tune calibration for CH3 V; <2:0> for the combiner input and <4:3> for the combiner output; <7:5> reserved bits;
Quad0	d2a_CH0_extra	4	yes	no	<0> for the CH0 B0 Wilkinson; <1> for the CH0 B1 Wilkinson; <3> for reserved bit;	reserved bits
Quad1	d2a_CH1_extra	4	yes	no	<0> for the CH1 B0 Wilkinson; <1> for the CH1 B1 Wilkinson; <3> for reserved bit;	reserved bits
Quad2	d2a_CH2_extra	4	yes	no	<1> for the CH2 B1 Wilkinson; <3> for reserved bit;	reserved bits
Quad3	d2a_CH3_extra	4	yes	no	<0> for the CH3 B0 Wilkinson; <1> for the CH3 B1 Wilkinson; <3> for reserved bit;	reserved bits
Center	d2a_extra_bits_B0	8	yes	yes	Capacitance tune calibration for DIST combiners; <7:4> for the DIST St1 combiners for beam0; <3:0> reserved bits;	Capacitance tune calibration for center combiners; <7:6> for the St1 distribution network input at Beam 0; <0:5> for the bias of distribution network at Beam 0;
Center	d2a_extra_bits_B1	8	yes	yes	Capacitance tune calibration for DIST combiners; <7:4> for the DIST St1 combiners for beam1; <3:0> reserved bits;	Capacitance tune calibration for center combiners; <7:6> for the St1 distribution network input at Beam 1; <0:5> for the bias of distribution network at Beam 1;
Center	d2a_extra_bits_B2	8	yes	yes	<4:0> for d2a_bias_adjust<4:0>; <5> for d2a_slope_2x_en; <7:6> reserved bits;	<4:0> for d2a_bias_adjust<4:0>; <5> for d2a_slope_2x_en; <7:6> reserved bits;
Center	d2a_extra_bits_misc	16	No	yes	reserved bits	Capacitance tune calibration for center combiners; <3:0> for the distribution network interstage at Beam<1:0>; <15:4> reserverd bits
Center	d2a_daisy_chain0_bias	6	yes	yes	reserved bits	reserved bits
Center	d2a_daisy_chain1_bias	6	yes	yes	reserved bits	reserved bits
Center	d2a_daisy_chain2_bias	6	yes	yes	reserved bits	reserved bits
Center	d2a_daisy_chain3_bias	6	yes	yes	reserved bits	reserved bits
Center	d2a_daisy_chain0_tuning	4	yes	no	reserved bits	reserved bits
Center	d2a_daisy_chain1_tuning	4	yes	no	reserved bits	reserved bits
Center	d2a_daisy_chain2_tuning	4	yes	no	reserved bits	reserved bits
Center	d2a_daisy_chain3_tuning	4	yes	no	reserved bits	reserved bits
Center	d2a_daisy_chain0_cbias	3	no	no	reserved bits	reserved bits
Center	d2a_daisy_chain1_cbias	3	no	no	reserved bits	reserved bits
Center	d2a_daisy_chain2_cbias	3	no	no	reserved bits	reserved bits

Center	d2a_daisy_chain3_cbias	3	no	no	reserved bits	reserved bits
Center	d2a_daisy_chain0_offsetbias	1	no	no	reserved bits	reserved bits
Center	d2a_daisy_chain1_offsetbias	1	no	no	reserved bits	reserved bits
Center	d2a_daisy_chain2_offsetbias	1	no	no	reserved bits	reserved bits
Center	d2a_daisy_chain3_offsetbias	1	no	no	reserved bits	reserved bits

Table 11: Bias and Tuning Register Definitions

DCTEST configuration

The DCTEST ADC allows for sampling of various DC values from around the chip. The signal captured by the ADC is determined by the source enable, as shown in Table 12, and the mode, in Table 13.

DCTEST EN	Blueway Block		Stampede Block	
DCTEST_EN_QUAD0<0>	Quad0 H	LNA St1 and St2	Quad0 V	PA and DRV
DCTEST_EN_QUAD0<1>	Quad0 H	Splitter St3	Quad0 V	Comb
DCTEST_EN_QUAD0<2>	Quad0 V	LNA St1 and St2	Quad0 H	PA and DRV
DCTEST_EN_QUAD0<3>	Quad0 V	Splitter St3	Quad0 H	Comb
DCTEST_EN_QUAD1<0>	Quad1 H	LNA St1 and St2	Quad1 V	PA and DRV
DCTEST_EN_QUAD1<1>	Quad1 H	Splitter St3	Quad1 V	Comb
DCTEST_EN_QUAD1<2>	Quad1 V	LNA St1 and St2	Quad1 H	PA and DRV
DCTEST_EN_QUAD1<3>	Quad1 V	Splitter St3	Quad1 H	Comb
DCTEST_EN_QUAD2<0>	Quad2 H	LNA St1 and St2	Quad2 V	PA and DRV
DCTEST_EN_QUAD2<1>	Quad2 H	Splitter St3	Quad2 V	Comb
DCTEST_EN_QUAD2<2>	Quad2 V	LNA St1 and St2	Quad2 H	PA and DRV
DCTEST_EN_QUAD2<3>	Quad2 V	Splitter St3	Quad2 H	Comb
DCTEST_EN_QUAD3<0>	Quad3 H	LNA St1 and St2	Quad3 V	PA and DRV
DCTEST_EN_QUAD3<1>	Quad3 H	Splitter St3	Quad3 V	Comb
DCTEST_EN_QUAD3<2>	Quad3 V	LNA St1 and St2	Quad3 H	PA and DRV
DCTEST_EN_QUAD3<3>	Quad3 V	Splitter St3	Quad3 H	Comb
DCTEST_EN_CENTER<0>	Center B0	DIST_St1	Center B0	DIST St1 and St2
DCTEST_EN_CENTER<1>	Center B1	DIST_St1	Center B0	DIST St1 and St2
DCTEST_EN_CENTER<2>	Center	Reserved bit	Center B1	DIST St1 and St2
DCTEST_EN_CENTER<3>	Center	Reserved bit	Center B1	DIST St1 and St2
DCTEST_EN_CENTER<4>	Center	Reserved bit	Center	Reserved bit
DCTEST_EN_CENTER<5>	Center	Reserved bit	Center	Reserved bit
DCTEST_EN_CENTER<6>	Center	Reserved bit	Center	Reserved bit
DCTEST_EN_CENTER<7>	Center	Reserved bit	Center	Reserved bit
DCTEST_EN_CENTER<8>	Center	Reserved bit	Center	Reserved bit
DCTEST_EN_CENTER<9>	Center	Global PTAT Bias	Center	Global PTAT Bias

Table 12: DCTEST Source

Blueway Source	Blueway Modes (DCTEST_MODE)	Stampede Source	Stampede Modes (DCTEST_MODES)
LNA St1 and St2	111 => vddby2_test_St1; 110=> vddby2_test_St2; 101=> GND; 100=> VDD_1p8V_res; 011=> IB_1st_LNA; 010=> IB_2nd_LNA; 001=> GND; 000=> GND;	PA St2 and St3	111 => VDD_DRV_HF; 110=> VDD_PA_QT; 101=> VG_PA<0>; 100=> VG_PA<1>; 011=> VG_DRV<0>; 010=> VG_DRV<1>; 001=> GND; 000=> GND;
Splitter St3	111 => vddby2_test; 110=> IB_LNA; 101=> CG1_Beam0; 100=> GND; 011=> CG1_Beam1; 010=> GND; 001=> IB_LNA; 000=> GND;	Comb St1	111 => vddby2_test; 110=> VG2s<2>; 101=> VG2s<1>; 100=> VG2s<0>; 011=> VG1m<2>; 010=> VG1m<1>; 001=> VG1m<0>; 000=> GND;
DIST_St1	111 => vddby2_test; 110=> GND; 101=> VB0; 100=> IB_Comb; 011=> VG1_CG1_CH1; 010=> VG2_CG1_CH2; 001=> GND; 000=> VC_CG2_Diode;	DIST St1 and St2	111 => vddby2_test; 110=> Gbias1_stg2<3>; 101=> Gbias1_stg2<2>; 100=> Gbias1_stg2<1>; 011=> Gbias1_stg2<0>; 010=> Gbias_mux<2>; 001=> Gbias_mux<1>; 000=> Gbias_mux<0>;
DIST_St2_0	Reserved bit due to passive Wilkinson	DIST St1 and St2	111 => GND; 110=> GND 101=> GND 100=> GND 011=> GND 010=> GND; 001=> Cbias1_stg1<1>; 000=> Cbias1_stg1<0>;

DIST_St2_1	Reserved bit due to passive Wilkinson		

Table 13: DCTEST Modes

SRAM Beam Table

The beam tables storage is implemented with a SRAM IP instantiated in each Quad. The SRAM has a single read and write port and stores the table settings for both the polarizations of RX and TX patterns. Thus, when the beam table index is updated the SRAM needs to be read and the output stored in a holding register. The architecture of this is shown in the figure below.

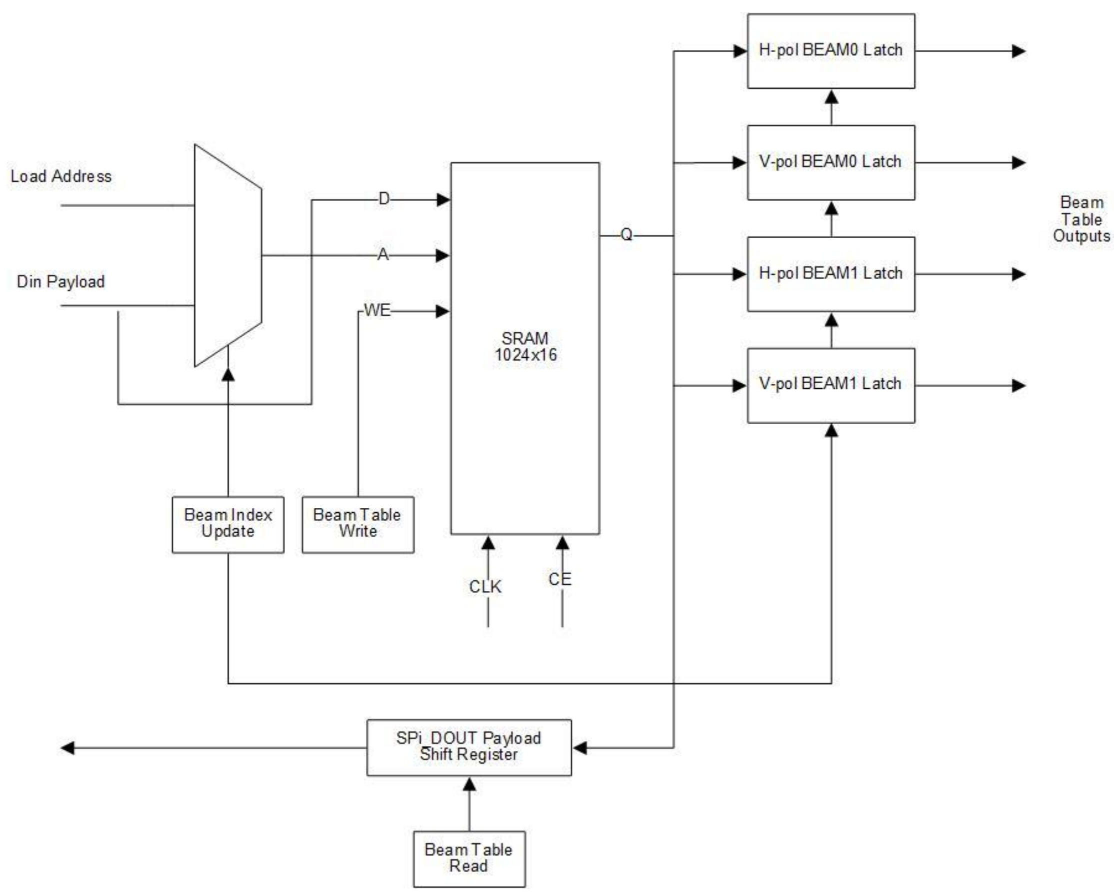


Figure 22: SRAM beam table architecture

The beam pointers for accessing data from the SRAM are local to the QUAD, but the address for loading of the SRAM is global across the chip. To map from the SPI address of the BEAM TABLE to the BEAM POINTER use FLOOR(BEAM_TABLE_ADDRESS/4). For example, see the table below.

QUAD 0 SPI Address	QUAD 1 SPI Address	QUAD 2 SPI Address	QUAD 3 SPI Address	Beam Pointer Value
0x000	0x001	0x002	0x003	0x000
0x004	0x005	0x006	0x007	0x001
…	…	…	…	…
0xFFC	0xFFD	0xFFE	0xFFF	0x3FF

Table 14: SRAM address to Beam pointer mapping

The SRAM is common to all BEAM formers in a QUAD so if independent beam control sets are desired for each beam then the pointers should use independent sections of the SRAM.

Initialization and Programming Examples (To Be Added)